

APTRANSCO AEE (CIVIL) REVISION KIT

DESIGN OF REINFORCED CONCRETE & CONCRETE TECHNOLOGY (DDRC)

PART 1: TOP 60 GOLDEN FACTS

1. Reinforced concrete design relies on the assumption of a perfect strain bond between steel and concrete at their interface.
2. The characteristic strength f_{ck} is defined as the stress below which not more than 5% of the test results are expected to fall.
3. Concrete characteristic strength is determined by testing standard 150 mm cubes after curing for exactly 28 days.
4. Apparent compressive strength of concrete is lower in standard cylinders (150 mm diameter $\times$ 300 mm height) than standard cubes ($f_{cylinder} \approx 0.80 \cdot f_{cube}$).
5. The flexural tensile strength of concrete (modulus of rupture) is represented empirically by $f_{cr} = 0.7 \cdot \sqrt{f_{ck}} \text{ N/mm}^2$.
6. Short-term static modulus of elasticity of concrete is given by the IS 456-2000 relation $E_c = 5000 \cdot \sqrt{f_{ck}} \text{ N/mm}^2$.
7. Long-term static modulus of elasticity of concrete accounts for creep through the effective modulus $E_{ce} = E_c / (1 + \theta)$.
8. Creep coefficient (θ) values decrease with loading age, dropping from 2.2 at 7 days to 1.6 at 28 days and 1.1 at 1 year.
9. Soundness of cement due to excess lime is measured on-site using the standard Le Chatelier split-ring cylinder apparatus.
10. Soundness of cement due to magnesia is measured using the autoclave test under high steam pressure.
11. Standard consistency of cement paste is determined using a 10 mm diameter Vicat plunger penetrating 5 to 7 mm from the bottom.
12. Water requirement for the standard initial setting time test is taken as $0.85 \cdot P$, where P is the standard consistency.
13. Hand-mixed concrete on-site requires a mandatory 10% increase in cement content to offset lower mixing efficiency.
14. Under Working Stress Design (WSD), concrete and steel are assumed to behave elastically under service loads.
15. Limit State Design (LSD) utilizes partial safety factors for loads (γ_f) and material strengths (γ_m) to assess safety at failure.
16. The partial safety factor for concrete in LSD is $\gamma_{mc} = 1.50$ to account for higher variation in field casting.
17. The partial safety factor for steel in LSD is $\gamma_{ms} = 1.15$ to reflect more controlled industrial manufacturing conditions.
18. In Limit State flexure, the maximum compressive strain in concrete at the outermost compression fiber is limited to 0.0035.

19. Compressive stress block of concrete in LSD is modeled as a combination of parabolic and rectangular stress zones.
20. The depth of the parabolic zone of the LSD stress block is $0.57 \cdot x_u$, while the rectangular zone is $0.43 \cdot x_u$.
21. At the tension reinforcement level, the minimum strain at failure must not be less than $0.002 + 0.87 \cdot f_y / E_s$.
22. An under-reinforced section fails through steel yielding first, providing ample visual cracking warning before ultimate collapse.
23. An over-reinforced section fails through sudden concrete crushing without warning and is strictly prohibited by IS 456-2000.
24. In balanced sections, both concrete and steel reach their design strain limits simultaneously under ultimate load.
25. Slabs are classified as one-way systems when the aspect ratio of the longer span to the shorter span (L_y / L_x) is greater than 2.0.
26. Slabs are classified as two-way systems when the aspect ratio (L_y / L_x) is less than or equal to 2.0.
27. Slabs monolithic with edge beams produce compatibility torsion, which allows torsional stiffness to be neglected in the design.
28. Slabs monolithic with interior beams are designed for equilibrium torsion, requiring full torsional reinforcement.
29. Flat slabs are reinforced concrete floors supported directly by columns without intermediate beams.
30. Drop panels are localized thickenings of the flat slab around column heads to enhance shear and punching capacity.
31. Column capitals are flared funnel-shaped column heads designed to reduce the effective span and shear stresses in flat slabs.
32. The critical section for isolated footing bending moment is located at the physical face of the column.
33. The critical section for isolated footing one-way shear is located at a distance of effective depth d from the column face.
34. The critical section for isolated footing two-way (punching) shear is located at a distance of $d/2$ from the column face.
35. Shear reinforcement stirrups subjected to torsion must be provided as closed loops perpendicular to the beam axis.
36. Bent-up bars are permitted to act as shear reinforcement but are restricted from carrying more than 50% of the total design shear.
37. Nominal shear stress τ_v in T-beams is computed using only the web width b_w rather than the flange width.
38. Development length L_d represents the minimum embedment depth required to transmit the design stress from steel to concrete.
39. Pull-out tests for bond stress evaluate the reinforcing bar in tension while compressing the surrounding concrete block.
40. Deformed bars (HYSD) have surface ribs that mechanically anchor into concrete, raising bond strength by 60% over plain bars.
41. Design bond stress τ_{bd} is increased by 25% when the reinforcing bar is placed in a zone of structural compression.

42. Tension hooks and bends are highly effective anchors, but they lose all anchorage value when placed in compression zones.
43. Lap splices are used to transfer forces between different bars and must be staggered to avoid localized stress concentrations.
44. Tension lap splices must not be used for reinforcing bars exceeding 32 mm in diameter unless welded on-site.
45. In beams subjected to torsion, the threshold depth for providing side-face reinforcement drops from 750 mm to 450 mm.
46. Side-face reinforcement must be distributed equally on both vertical faces of a deep beam.
47. Column longitudinal reinforcement must lie between 0.8% and 6.0% of the gross cross-sectional concrete area.
48. In columns, the maximum longitudinal reinforcement is practically capped at 4.0% to avoid steel congestion during site-lapping.
49. Column lateral ties prevent premature buckling of the longitudinal steel and contain lateral bursting stresses.
50. Helical spiral reinforcement provides continuous lateral confinement, raising circular column axial capacity by 5%.
51. The minimum eccentricity required for column design is 20 mm to account for accidental construction misalignments.
52. Footing design calculations for bending moment and shear must strictly omit the self-weight of the footing.
53. Footing self-weight is only included when sizing the total base area against the Soil Bearing Capacity (SBC).
54. Continuous concrete beams have points of inflection where the bending moment curve changes sign from tension to compression.
55. Diagonal cracks in concrete members under pure torsion propagate in a continuous spiral pattern at 45° around the beam axis.
56. Deep foundations like driven piles are evaluated using a deep base bearing capacity factor $N_c = 9.0$ under cohesive conditions.
57. Sand replacement tests are used on-site to determine the dry density of granular sands where core-cutters fail.
58. Compaction factor tests measure the workability of low-water concrete mixes where standard slump tests are insensitive.
59. The compaction curve of clayey soils exhibits an Optimum Moisture Content (OMC) corresponding to Maximum Dry Density.
60. It is physically impossible to compact any soil beyond the theoretical Zero Air Voids (ZAV) boundary line on-site.

PART 2: TOP 50 GOVERNING FORMULAS

1. Modulus of Elasticity of Concrete: $E_c = 5000 \cdot \sqrt{f_{ck}} \text{ N/mm}^2$
2. Effective Modulus of Concrete (Creep): $E_{ce} = E_c / (1 + \theta)$
3. Modular Ratio under Working Stress Design: $m = 280 / (3 \cdot \sigma_{cbc})$
4. Balanced Neutral Axis Depth Factor (WSD): $k = m \cdot \sigma_{cbc} / (m \cdot \sigma_{cbc} + \sigma_{st})$
5. Balanced Moment of Resistance Constant (WSD): $Q = 0.5 \cdot \sigma_{cbc} \cdot j \cdot k$

6. Limiting Neutral Axis Depth Ratio (Fe250): $x_{u,max} / d = 0.53$
7. Limiting Neutral Axis Depth Ratio (Fe415): $x_{u,max} / d = 0.48$
8. Limiting Neutral Axis Depth Ratio (Fe500): $x_{u,max} / d = 0.46$
9. Balanced Moment of Resistance (Fe250): $M_{u,lim} = 0.148 \cdot f_{ck} \cdot b \cdot d^2$
10. Balanced Moment of Resistance (Fe415): $M_{u,lim} = 0.138 \cdot f_{ck} \cdot b \cdot d^2$
11. Balanced Moment of Resistance (Fe500): $M_{u,lim} = 0.133 \cdot f_{ck} \cdot b \cdot d^2$
12. Characteristic Flexural Tensile Strength: $f_{cr} = 0.7 \cdot \sqrt{f_{ck}}$
13. Splitting Tensile Strength: $f_{sp} = 2 \cdot P / (\pi \cdot L \cdot D)$
14. Depth of Neutral Axis (LSD Singly): $x_u = (0.87 \cdot f_y \cdot A_{st}) / (0.36 \cdot f_{ck} \cdot b)$
15. Ultimate Moment of Resistance (Singly): $M_u = 0.87 \cdot f_y \cdot A_{st} \cdot d \cdot [1 - (f_y \cdot A_{st}) / (f_{ck} \cdot b \cdot d)]$
16. Minimum Steel Area in Beams: $A_{s,min} / (b \cdot d) = 0.85 / f_y$
17. Maximum Steel Area in Beams (Tension or Compression): $A_{s,max} = 0.04 \cdot b \cdot d$
18. Nominal Shear Stress in Beams: $\tau_v = V_u / (b \cdot d)$
19. Nominal Shear Stress in flanged T-beams: $\tau_v = V_u / (b_w \cdot d)$
20. Minimum Area of Shear Stirrups: $A_{sv} / (b \cdot s_v) \geq 0.4 / (0.87 \cdot f_y)$
21. Maximum Spacing of Vertical Shear Stirrups: $s_{v,max} = 0.75 \cdot d \leq 300 \text{ mm}$
22. Ultimate Shear Capacity of Concrete (Unreinforced Web): $V_{uc} = \tau_c \cdot b \cdot d$
23. Design Shear Capacity of Vertical Stirrups: $V_{us} = (0.87 \cdot f_y \cdot A_{sv} \cdot d) / s_v$
24. Equivalent Shear Force (Torsional Beam): $V_e = V_u + 1.6 \cdot T_u / b$
25. Equivalent Nominal Shear Stress (Torsion): $\tau_{ve} = V_e / (b \cdot d)$
26. Equivalent Bending Moment (Torsion): $M_{e1} = M_u + M_t$
27. Torsional Moment Contribution: $M_t = T_u \cdot [1 + D / b] / 1.7$
28. Equivalent Compression Bending Moment: $M_{e2} = M_t - M_u$
29. Basic Development Length of Reinforcing Bar: $L_d = (\phi \cdot \sigma_s) / (4 \cdot \tau_{bd})$
30. Development Length (Deformed tension): $L_d = (\phi \cdot \sigma_s) / (4 \cdot 1.6 \cdot \tau_{bd})$
31. Development Length (Deformed compression): $L_d = (\phi \cdot \sigma_s) / (4 \cdot 1.6 \cdot 1.25 \cdot \tau_{bd}) = (\phi \cdot \sigma_s) / (8 \cdot \tau_{bd})$
32. Development Length Check at Simple Supports: $L_d \leq 1.3 \cdot M_1 / V + L_0$
33. Development Length Check at Points of Inflection: $L_d \leq M_1 / V + L_0$
34. Anchorage Value of 90-degree Bend: $L_a = 8 \cdot \phi$
35. Anchorage Value of 180-degree Hook: $L_a = 16 \cdot \phi$
36. Side Face Reinforcement Area (Torsion/Deep Beams): $A_{sf} = 0.001 \cdot b_w \cdot D$
37. Maximum Spacing of Side Face Reinforcement: $s_{max} = 300 \text{ mm}$
38. Ultimate Axial Load Capacity (Tied Columns): $P_u = 0.4 \cdot f_{ck} \cdot A_c + 0.67 \cdot f_y \cdot A_{sc}$
39. Net Concrete Core Area in Columns: $A_c = A_g - A_{sc}$
40. Ultimate Axial Capacity of Helical Circular Columns: $P_{u,helical} = 1.05 \cdot [0.4 \cdot f_{ck} \cdot A_c + 0.67 \cdot f_y \cdot A_{sc}]$

41. Spiral Volumetric Reinforcement Ratio: $V_{sp} / V_c \geq 0.36 \cdot [A_g / A_c - 1] \cdot f_{ck} / f_y$
42. Column Minimum Eccentricity: $e_{min} = L / 500 + D / 30 \geq 20 \text{ mm}$
43. Maximum Diameter of lateral ties in columns: $d_{tie} \geq \max[d_{long,max} / 4, 6 \text{ mm}]$
44. Maximum Spacing of lateral ties in columns: $s_{tie} = \min[\text{Least Lateral Dimension}, 16 \cdot d_{long,min}, 300 \text{ mm}]$
45. Maximum Diameter of Main Slab Bars: $d_{bar} \leq D_{slab} / 8$
46. Main Tension Reinforcement Spacing (Slabs): $s_{main} \leq \min[3 \cdot d, 300 \text{ mm}]$
47. Distribution Reinforcement Spacing (Slabs): $s_{dist} \leq \min[5 \cdot d, 450 \text{ mm}]$
48. Isolated Footing Minimum Thickness at Edges (Soil): $t_{edge} \geq 150 \text{ mm}$
49. Isolated Footing Minimum Thickness at Edges (Piles): $t_{edge} \geq 300 \text{ mm}$
50. Characteristic Tensile Strength (Split tension): $f_t = 0.7 \cdot \sqrt{f_{ck}}$

PART 3: TOP 50 STANDARD VALUES & LIMITS

1. Permissible bending compressive stress σ_{cbc} for M20 concrete (WSM): 7.0 N/mm^2
2. Permissible bending compressive stress σ_{cbc} for M25 concrete (WSM): 8.5 N/mm^2
3. Permissible bending compressive stress σ_{cbc} for M30 concrete (WSM): 10.0 N/mm^2
4. Modular ratio m for M20 concrete under WSM rules: 13.33
5. Modular ratio m for M25 concrete under WSM rules: 10.98
6. Permissible steel tensile stress σ_{st} for Fe415 (WSM): 230 N/mm^2
7. Permissible steel tensile stress σ_{st} for Fe250 (WSM): 140 N/mm^2
8. Minimum concrete grade for reinforced concrete under mild exposure: M20
9. Minimum concrete grade for reinforced concrete under moderate exposure: M25
10. Minimum concrete grade for reinforced concrete under severe exposure: M30
11. Design bond stress τ_{bd} for M20 concrete (Plain tension): 1.2 N/mm^2
12. Design bond stress τ_{bd} for M25 concrete (Plain tension): 1.4 N/mm^2
13. Design bond stress τ_{bd} for M30 concrete (Plain tension): 1.5 N/mm^2
14. Design bond stress τ_{bd} for M35 concrete (Plain tension): 1.7 N/mm^2
15. Design bond stress τ_{bd} for M40 concrete (Plain tension): 1.9 N/mm^2
16. Maximum design shear stress $\tau_{c,max}$ for M20 concrete: 2.8 N/mm^2
17. Maximum design shear stress $\tau_{c,max}$ for M25 concrete: 3.1 N/mm^2
18. Maximum design shear stress $\tau_{c,max}$ for M30 concrete: 3.5 N/mm^2
19. Maximum design shear stress $\tau_{c,max}$ for M35 concrete: 3.7 N/mm^2
20. Maximum design shear stress $\tau_{c,max}$ for M40 concrete: 4.0 N/mm^2
21. Minimum reinforcement area in slabs using Fe415 (HYSD): 0.12% of Gross Area
22. Minimum reinforcement area in slabs using Fe250 (Mild Steel): 0.15% of Gross Area
23. Maximum size of aggregates in standard concrete structural components: 20 mm

24. Nominal clear cover to reinforcement in concrete slabs: 20 mm
25. Nominal clear cover to reinforcement in concrete beams: 25 mm
26. Nominal clear cover to reinforcement in concrete columns: 40 mm
27. Nominal clear cover to reinforcement in concrete foundations/footings: 50 mm
28. Permissible individual variation of cube compressive tests: $\pm 15\%$ of Average
29. Standard curing temperature of concrete test cubes: 27 ± 2 degrees Celsius
30. Curing relative humidity requirement for concrete test cubes: $\geq 90\%$
31. Time of prop removal for concrete slabs spanning up to 4.5 m: 7 days
32. Time of prop removal for concrete slabs spanning over 4.5 m: 14 days
33. Time of prop removal for concrete beams spanning up to 6.0 m: 14 days
34. Time of prop removal for concrete beams spanning over 6.0 m: 21 days
35. Time of formwork stripping for beam vertical sides and columns: 16 to 24 hours
36. Minimum longitudinal column bar diameter: 12 mm
37. Minimum number of longitudinal bars in a rectangular column: 4
38. Minimum number of longitudinal bars in a circular column: 6
39. Minimum longitudinal steel percentage in concrete columns: 0.8% of Gross Area
40. Maximum longitudinal steel percentage in concrete columns: 6.0% of Gross Area
41. Maximum longitudinal steel percentage in columns under standard site-lapping: 4.0%
42. Nominal clear cover reduction for columns of diameter ≤ 200 mm with bars ≤ 12 mm: 25 mm
43. Standard concrete cube specimen dimensions: 150 mm $\times$ 150 mm $\times$ 150 mm
44. Standard concrete cylinder specimen dimensions: 150 mm diameter $\times$ 300 mm height
45. Ratio of tensile splitting strength to compressive strength of concrete: 10% to 15%
46. Coefficient of thermal expansion of concrete (approximate match to steel): $11 \cdot 10^{-6}$ per degree C
47. Mass density of plain concrete (PCC): 2400 kg/m³ (or 24 kN/m³)
48. Mass density of reinforced concrete (RCC): 2500 kg/m³ (or 25 kN/m³)
49. Standard slump range for pumpable concrete mixes: 75 mm to 100 mm
50. Standard slump range for concrete in heavily reinforced sections: 50 mm to 100 mm

PART 4: TOP 50 RECURRING EXAMINER TRAPS

1. Swapping Gross area with Net concrete area ($A_c = A_g - A_{sc}$) in column capacity checks.
2. Forgetting the deformed bar factor of 1.6 when computing development lengths in tension.
3. Forgetting that concrete hand-mixing requires a 10% penalty increase in cement content on-site.
4. Using the overall beam depth D instead of the effective depth d in nominal shear calculations.
5. Assuming hooks provide anchorage in compression zone steel (anchorage value is exactly zero).
6. Swapping the footing critical offsets: checking punching shear at d and one-way shear at d/2.

7. Direct averaging of three cube test strengths where one specimen violates the $\pm 15\%$ variation limit.
8. Swapping the 1.05 circular spiral column capacity multiplier with tied columns.
9. Applying the 1.05 helical column capacity multiplier without verifying the volumetric spiral ratio limit.
10. Using $5700 \cdot \sqrt{f_{ck}}$ from the superseded IS 456-1978 code instead of $5000 \cdot \sqrt{f_{ck}}$.
11. Confusing concrete characteristic strength f_{ck} with WSM permissible bending stress σ_{cbc} .
12. Confusing Modulus of Rupture ($0.7 \cdot \sqrt{f_{ck}}$) with the true direct tensile strength ($0.35 \cdot \sqrt{f_{ck}}$).
13. Assuming that raising carbon content in steel increases its elastic modulus E_s (which remains constant).
14. Swapping the 0.12% slab steel limit (HYSD) with the 0.15% slab steel limit (mild steel).
15. Splicing reinforcing bars larger than 32 mm in tension without welding them on-site.
16. Using the larger bar diameter instead of the smaller bar diameter when calculating lap lengths of splices.
17. Confusing slab concrete cover rules with beam concrete cover rules during detailing.
18. Forgetting the 1.30 multiplier for M_1 / V at simple compressive supports.
19. Using the overall slab thickness D instead of effective depth d to check maximum spacing limits.
20. Swapping the flat slab capital funnel head definition with the drop panel thickened slab section.
21. Using the factored load instead of the service load when sizing foundation base areas.
22. Swapping the $0.25 \cdot f_{ck}$ WSD pedestal bearing limit with the $0.45 \cdot f_{ck}$ limit of Limit State Design.
23. Swapping aggregate testing standards (IS 2386) with fresh concrete testing standards (IS 1199).
24. Using the overall beam depth D in the M_t torsion geometric ratio instead of the effective depth d .
25. Forgetting that side-face reinforcement thresholds drop to 450 mm from 750 mm under torsional moments.
26. Sizing shear stirrup spacing using the overall beam depth D instead of the effective depth d .
27. Splicing reinforcing bars at the location of maximum bending moment instead of low stress regions.
28. Using the nominal bar diameter instead of the gross diameter in rivet joint spacing limits.
29. Swapping the $3d$ main steel slab spacing limit with the $5d$ secondary steel slab limit.
30. Using the gross diameter of a pile to determine the end conditions instead of the core diameter.
31. Applying the $0.4 \cdot f_{ck} \cdot A_c + 0.67 \cdot f_y \cdot A_{sc}$ formula for columns when eccentricity exceeds $0.05 \cdot D$.
32. Confusing the compaction factor with the compaction dry density under Geotechnical testing.
33. Assuming that a flexible footing has non-uniform soil pressure instead of uniform soil pressure.
34. Swapping the critical sections for checking punching shear under shallow footings vs. pile foundations.
35. Omitting the self-weight of the footing when evaluating critical shear and bending stresses in the pad.
36. Using the unconfined compression strength q_u directly as cohesion c_u (cohesion is actually $q_u / 2$).
37. Forgetting that sheepfoot rollers compact cohesive clays through kneading action from the bottom-up.
38. Swapping the definitions of one-way slabs with two-way slabs when the aspect ratio equals exactly 2.0.
39. Swapping the time limits for prop removal under slabs vs. beams.
40. Using 2500 kg/m^3 as the density of plain concrete, which belongs strictly to reinforced concrete (PCC is 2400).
41. Swapping the critical sections of footings under masonry walls (midway between center and edge) with concrete columns.

42. Forgetting that the neutral axis of a flanged T-beam lies in the web when the flange thickness is very small.
43. Using the nominal diameter of a reinforcing bar in the L_d check when it has been reduced by corrosion on-site.
44. Applying the 1.30 development length bonus at a point of inflection (it is strictly limited to simple supports).
45. Assuming that aggregate impact value measures hardness instead of toughness under mechanical load.
46. Swapping the Le Chatelier split cylinder test (free lime) with the autoclave high steam test (magnesia).
47. Confusing the initial setting time of cement (minimum 30 minutes) with the final setting time (maximum 10 hours).
48. Splicing column bars at the floor level instead of the mid-height zone of low bending stress.
49. Omitting the concrete edge cover casing on-site, leaving steel reinforcement vulnerable to direct soil moisture.
50. Confusing the active pressure coefficient K_a with the passive pressure coefficient K_p in retaining wall calculations.

PART 5: TOP 25 'NEVER FORGET' CONCEPTS

1. Modulus of elasticity of concrete is a function of compressive strength and is represented by $E_c = 5000 \cdot \sqrt{f_{ck}}$.
2. Hooke's Law is valid only up to the proportional limit of the material, not the elastic limit.
3. For standard structural concrete and steel, modular ratio under WSM is strictly $m = 280 / (3 \cdot \sigma_{cbc})$.
4. Deformed HYSD bars in tension require a 60% boost in design bond stress over plain bars.
5. Splicing bars of different diameters must be calculated using the diameter of the smaller bar.
6. The maximum spacing of vertical shear stirrups is capped at $0.75 \cdot d$ or 300 mm.
7. Side-face reinforcement is required if the depth of a torsional beam exceeds 450 mm.
8. The critical section for isolated footing bending moment is located at the face of the column.
9. Punching shear in footings acts along a critical perimeter located at $d/2$ from the column face.
10. One-way shear in footings acts along a critical section located at distance d from the column face.
11. Column longitudinal reinforcement must remain between 0.8% and 6.0% of the gross area.
12. Helical circular columns are allowed a 5% capacity increase if the volumetric spiral ratio limit is satisfied.
13. Maximum steel area in reinforced concrete beams is strictly limited to 4% of the gross area to prevent crowding.
14. Under IS 456-2000, standard hooks and bends have exactly zero anchorage value in compression zone steel.
15. Concrete test results are valid only if individual cube strengths vary by no more than $\pm 15\%$ of the average.
16. Slabs with aspect ratios $L_y / L_x > 2.0$ behave structurally as one-way spanning systems.
17. Slabs with aspect ratios $L_y / L_x \leq 2.0$ behave structurally as two-way spanning systems.
18. Drop panels and column capitals are used in flat slabs to mitigate punching shear failure modes.
19. Standard cylinder compressive strength is taken as approximately 80% of the standard cube strength.
20. Compaction factor tests are used for dry, low-workability concrete mixes where slump values are zero.
21. Hand-mixed concrete requires a mandatory 10% increase in cement content to maintain characteristic strength.

22. The partial safety factor for concrete is 1.50, while steel is 1.15 under Limit State Design.
23. The maximum strain in the outermost concrete compression fiber under LSD flexure is limited to 0.0035.
24. Over-reinforced beams are prohibited by IS 456-2000 because they exhibit sudden, explosive compression failure.
25. Zero Air Voids (ZAV) represents the absolute physical boundary limit of compaction for any soil on-site.

DESIGN OF STEEL STRUCTURES

APTRANSCO AEE (CIVIL) HIGH-YIELD FINAL REVISION EXAM KIT

PART 1: TOP 60 GOLDEN FACTS

1. Structural steel is categorized as a ductile material, showing a distinct yield point and a significant plastic range prior to fracture.
2. The yield strength of structural steel is the primary mechanical property used to identify steel grades and designate sections.
3. According to IS 2062, structural steel has a maximum carbon content of 0.23% to prevent weld zone embrittlement and preserve ductility.
4. Rolled steel sections are designated by their depth and weight per unit length, e.g., ISMB 400 @ 61.5 kg/m.
5. In rolled I-beams, the flanges resist major bending moments, while the web resists over 95% of the total transverse shear force.
6. Web buckling is a failure of the web of a beam acting as a column under heavy concentrated vertical loads at supports or load points.
7. Web crippling is the localized crushing of the web of a beam near the junction of the web and flange due to high compressive stress concentrations.
8. Web crippling can be prevented by providing a sufficient length of bearing or using bearing stiffeners at concentrated load locations.
9. Compound (built-up) beams are fabricated by adding cover plates to the flanges of rolled sections to increase their section modulus.
10. Symmetrical built-up sections are preferred over unsymmetrical ones to keep the centroid at mid-depth and avoid asymmetrical bending.
11. Cover plates of built-up beams are curtailed at sections where the design bending moment is less than the capacity of the un-curtailed section.
12. Structural connections are classified as rigid, semi-rigid, or flexible based on their moment-rotation capacity.
13. Riveted joints transfer loads through shear on the rivet shank and bearing between the rivet and the connected plates.
14. Unwin's formula relates the nominal diameter of a rivet (d) to the plate thickness (t): $d = 6.01 \cdot \sqrt{t}$ (in mm).
15. The gross diameter of a rivet hole is larger than the nominal diameter to allow for rivet expansion during driving.
16. Shearing failure of a rivet occurs when the transverse shear force exceeds the shear capacity of the rivet shank.
17. Bearing failure occurs when the plate material crushes against the rivet shank due to high contact stress.
18. Tensile tearing failure of a plate occurs across a row of rivet holes at the net sectional area of the plate.
19. Joint efficiency is the ratio of the minimum strength of the joint (least of shearing, bearing, tearing) to the strength of the solid plate.
20. Countersunk rivets are assumed to have a 33.3% reduction in tensile tearing strength compared to snap-headed rivets.
21. Welded joints offer higher efficiency, structural rigidity, and material savings compared to riveted or bolted joints.
22. Fillet welds have a triangular cross-section and are subjected primarily to shear stress on their throat plane.
23. The effective throat thickness (t_e) of a fillet weld is the minimum dimension from the root to the face, given by $t_e = k \cdot s$.
24. The fusion constant (k) depends on the angle between fusion faces, ranging from 0.70 for $60^\circ - 90^\circ$ to 0.50 for $114^\circ - 120^\circ$.
25. Butt (groove) welds are preferred for tension connections and are designed for direct tensile or compressive stresses.
26. Single-V and double-V butt welds must have full penetration to be considered 100% efficient in design.
27. Combined stresses in fillet welds are checked using the equivalent stress criterion, combining shear and normal stresses.
28. Unsymmetric weld groups are designed by balancing the lengths of weld runs to ensure the weld group centroid aligns with the gravity axis of the connected member.
29. Eccentric framed connections are subjected to combined direct shear force and a twisting moment about the connection centroid.
30. The elastic method assumes that shear stress is proportional to the distance from the connection centroid.

31. The critical bolt/rivet in an eccentric connection is the one that has the maximum vector sum of direct and twisting shear stresses.
32. The instantaneous center of rotation method is a plastic analysis method that considers the non-linear force-deformation behavior of fasteners.
33. Seated connections transfer loads primarily through bearing on a seat angle, with a top cleat angle provided to prevent tilting.
34. Simple columns are designed as axially loaded compression members using buckling curves as per IS 800-2007.
35. The slenderness ratio (λ) is the ratio of the effective length of the column to its minimum radius of gyration: $\lambda = L_e / r_{\min}$.
36. Long columns fail by sudden elastic buckling, whereas short columns fail by material crushing and yielding.
37. Symmetrical sections (like ISHB) have high radii of gyration about both axes, making them highly efficient as columns.
38. Compound columns are built up by connecting standard rolled sections with lacing systems or batten plates to increase buckling capacity.
39. Lacing systems are designed as pin-jointed trusses to resist a transverse shear force equal to 2.5% of the column's axial load.
40. Lacing bars are inclined at an angle between 40° and 70° to the longitudinal column axis.
41. Batten plates are designed as rigid frames subjected to combined bending and shear due to the column's transverse shear force.
42. Battened columns are penalized with a 10% increase in design slenderness ratio to account for shear deformation.
43. Columns bases are designed to distribute the heavy vertical column load over a larger area of the concrete pedestal.
44. Slab bases are flat steel plates placed under column flanges to distribute axial loads over concrete foundations.
45. Gusseted bases use gusset plates and angles to stiffen the base plate, allowing for a thinner plate design under heavy loads.
46. The maximum permissible bending stress in steel slab bases is limited to 1500 kg/cm² (150 MPa) under working stress design.
47. Grillage foundations consist of tiers of steel beams encased in concrete, used to support heavy column loads on weak soils.
48. Double-tier grillages use two perpendicular layers of steel beams to distribute loads in both orthogonal directions.
49. Roof trusses are triangulated frames designed to support roof coverings over large spans with minimal material.
50. Purlins are continuous horizontal beams spanning between roof trusses to support roof sheeting and resist wind loads.
51. The economic spacing of roof trusses is achieved when the cost of trusses equals the cost of purlins plus twice the cost of roof covering.
52. Wind loads on roof trusses are determined using pressure coefficients based on building height-to-width ratio and roof slope.
53. Plastic analysis is based on the ultimate limit state of collapse, where steel members form plastic hinges and undergo complete plastic yielding.
54. Plastic bending assumes that steel is an elastic-perfectly plastic material with constant yield stress post-yielding.
55. Plastic moment capacity (M_p) is the maximum moment a section can carry when it is fully plastic: $M_p = Z_p \cdot f_y$.
56. The shape factor (S) is the ratio of plastic section modulus (Z_p) to elastic section modulus (Z_e), representing structural reserve strength.
57. In fully plastic stage, the plastic neutral axis (PNA) divides the cross-section into two equal areas ($A_c = A_t = A/2$).
58. The lower bound (static) theorem states that any safe and statically admissible moment field yields a collapse load less than or equal to the true collapse load.
59. The upper bound (kinematic) theorem states that any kinematically admissible collapse mechanism yields a collapse load greater than or equal to the true collapse load.
60. The uniqueness theorem states that a collapse load is unique if it satisfies both equilibrium (lower bound) and mechanism (upper bound) conditions.

PART 2: TOP 50 GOVERNING FORMULAS

1. Young's Modulus and Shear Modulus Relation: $E = 2 \cdot G \cdot (1 + \nu)$
2. Young's Modulus and Bulk Modulus Relation: $E = 3 \cdot K \cdot (1 - 2\nu)$
3. Unwin's Rivet Sizing Formula: $d = 6.01 \cdot \sqrt{t}$
4. Gross Hole Diameter (for $d \leq 25$ mm): $d_g = d + 1.5$ mm

5. Gross Hole Diameter (for $d > 25$ mm): $d_g = d + 2.0$ mm
6. Rivet Shearing Capacity (Single Shear): $P_s = (\pi / 4) \cdot d_g^2 \cdot \tau_{vf}$
7. Rivet Shearing Capacity (Double Shear): $P_s = 2 \cdot (\pi / 4) \cdot d_g^2 \cdot \tau_{vf}$
8. Rivet Bearing Capacity: $P_b = d_g \cdot t \cdot \sigma_{pf}$
9. Plate Tensile Tearing Area Capacity: $P_t = (b - n \cdot d_g) \cdot t \cdot \sigma_{at}$
10. Rivet Joint Efficiency: $\eta = [\min(P_s, P_b, P_t) / P_{solid}] \cdot 100$
11. Solid Plate Tensile Capacity: $P_{solid} = b \cdot t \cdot \sigma_{at}$
12. Fillet Weld Throat Thickness: $t_t = k \cdot s$
13. Fillet Weld Design Strength (LSM): $f_{wd} = f_u / (\sqrt{3} \cdot \gamma_{mw})$
14. Butt Weld Design Strength (LSM, Tension): $P_{dw} = L_w \cdot t_t \cdot f_y / \gamma_{m0}$
15. Equivalent Weld Combined Stress: $f_e = \sqrt{f_a^2 + 3 \cdot q^2} \leq f_y / \gamma_{m0}$
16. Eccentric Connection Fastener Direct Force: $F_d = P / n$
17. Eccentric Connection Twisting Force: $F_m = (P \cdot e \cdot r) / \sum(r_i^2)$
18. Resultant Force in Eccentric Joint Fastener: $R = \sqrt{F_d^2 + F_m^2 + 2 \cdot F_d \cdot F_m \cdot \cos(\theta)}$
19. Radius of Gyration: $r = \sqrt{I / A}$
20. Column Slenderness Ratio: $\lambda = L_e / r_{min}$
21. Euler's Buckling Load: $P_{cr} = (\pi^2 \cdot E \cdot I) / L_e^2$
22. Perry-Robertson Design Compressive Stress: $f_{cd} = \chi \cdot f_y / \gamma_{m0}$
23. Modified Slenderness of Laced Columns: $\lambda_e = 1.05 \cdot \lambda$
24. Modified Slenderness of Battened Columns: $\lambda_e = 1.10 \cdot \lambda$
25. Transverse Shear Force for Column Lacing Design: $V_t = 0.025 \cdot P$
26. Base Plate Sizing Footprint Area: $A_{req} = P_{total} / q_{allow}$
27. Slab Base Thickness Sizing (IS 800-1984 WSM): $t = \sqrt{3 \cdot w \cdot (a^2 - b^2/4) / \sigma_{bs}}$
28. Slab Base Thickness Sizing (IS 800-2007 LSM): $t = \sqrt{2.5 \cdot w \cdot (a^2 - 0.3 \cdot b^2) \cdot \gamma_{m0} / f_y}$
29. Concrete Base Allowable Bearing Pressure (WSM): $q_{allow} = 0.25 \cdot f_{ck}$
30. Concrete Base Design Bearing Pressure (LSM): $q_{allow} = 0.45 \cdot f_{ck}$
31. Grillage Foundation Base Sizing: $A_{req} = (P + P_{self}) / SBC$
32. Purlin Wind Load Calculation: $F = (C_{pe} - C_{pi}) \cdot A \cdot q_p$
33. Purlin Allowable Live Load Reduction: $LL = 0.75 - 0.02 \cdot (\theta - 10^\circ)$
34. Elastic Section Modulus (Rectangle): $Z_e = (b \cdot d^2) / 6$
35. Plastic Section Modulus (Rectangle): $Z_p = (b \cdot d^2) / 4$
36. Elastic Section Modulus (Solid Circle): $Z_e = (\pi \cdot D^3) / 32$
37. Plastic Section Modulus (Solid Circle): $Z_p = D^3 / 6$
38. Shape Factor: $S = Z_p / Z_e$
39. Shape Factor of Rectangle: $S_{rect} = 1.50$
40. Shape Factor of Solid Circle: $S_{circ} = 16 / (3 \cdot \pi) \approx 1.70$
41. Shape Factor of Thin Hollow Tube: $S_{tube} = 4 / \pi \approx 1.27$
42. Shape Factor of Diamond Section: $S_{diamond} = 2.00$
43. Shape Factor of Isosceles Triangle: $S_{tri} = 2.34$
44. Symmetrical I-Section Shape Factor: $S_{I-beam} = 1.12$ to 1.15
45. Simply Supported Beam Collapse Load (Central Load): $W_c = 4 \cdot M_p / L$
46. Simply Supported Beam Collapse Load (UDL): $w_c = 8 \cdot M_p / L^2$
47. Fixed Beam Collapse Load (Central Load): $W_c = 8 \cdot M_p / L$
48. Fixed Beam Collapse Load (UDL): $w_c = 16 \cdot M_p / L^2$
49. Propped Cantilever Collapse Load (Central Load): $W_c = 6 \cdot M_p / L$
50. Propped Cantilever Collapse Load (UDL): $w_c = 11.656 \cdot M_p / L^2$

PART 3: TOP 50 STANDARD VALUES & LIMITS

1. Modulus of Elasticity of Steel (all grades): $2.0 \times 10^5 \text{ N/mm}^2$ (or **200 GPa**).
2. Modulus of Rigidity of Steel: $0.769 \times 10^5 \text{ N/mm}^2$ (or **76.9 GPa**).
3. Poisson's Ratio of Steel: **0.30** (elastic range) | **0.50** (plastic range).
4. Density of Structural Steel: **7850 kg/m³** (or **78.5 kN/m³**).
5. Coefficient of Linear Thermal Expansion of Steel: **$12 \times 10^{-6} / ^\circ\text{C}$** .
6. Design Yield Strength of structural steel Fe250: **250 MPa**.
7. Ultimate Tensile Strength of structural steel Fe250: **410 MPa**.
8. Maximum Carbon Content limit for weldable steel as per IS 2062: **0.23%**.
9. Standard Angle range between fusion faces for fillet welds: **60° to 120°**.
10. Weld Fusion Constant (k) for fusion angle 60° - 90°: **0.70**.
11. Weld Fusion Constant (k) for fusion angle 91° - 100°: **0.65**.
12. Weld Fusion Constant (k) for fusion angle 101° - 106°: **0.60**.
13. Weld Fusion Constant (k) for fusion angle 107° - 113°: **0.55**.
14. Weld Fusion Constant (k) for fusion angle 114° - 120°: **0.50**.
15. Minimum Size of Fillet Weld for plate thickness $\leq 10 \text{ mm}$: **3 mm**.
16. Minimum Size of Fillet Weld for plate thickness 10 mm - 20 mm: **5 mm**.
17. Minimum Size of Fillet Weld for plate thickness 20 mm - 32 mm: **6 mm**.
18. Minimum Size of Fillet Weld for plate thickness $> 32 \text{ mm}$: **8 mm**.
19. Maximum Size of Fillet Weld at square edge: **$t - 1.5 \text{ mm}$** .
20. Maximum Size of Fillet Weld at rounded toe: **$0.75 \cdot t$** .
21. Minimum Spacing (Pitch) of fasteners: **$2.5 \cdot d$** .
22. Maximum Pitch of fasteners in tension flange: **$16 \cdot t$ or **200 mm** (lesser).**
23. Maximum Pitch of fasteners in compression flange: **$12 \cdot t$ or **200 mm** (lesser).**
24. Maximum Pitch of tacking fasteners (plates back-to-back): **$32 \cdot t$ or **300 mm** (lesser).**
25. Minimum Edge Distance for hand-flame cut plate edges: **$1.7 \cdot d_0$** .
26. Minimum Edge Distance for machine-flame cut plate edges: **$1.5 \cdot d_0$** .
27. Angle of Inclination of lacing bars to the column axis: **40° to 70°**.
28. Maximum Slenderness Ratio (λ) of an individual lacing bar: **145**.
29. Maximum Slenderness Ratio for columns under dead and live loads: **180**.
30. Maximum Slenderness Ratio for columns under wind/seismic compression: **250**.
31. Maximum Slenderness Ratio for roof truss ties under wind reversal: **350**.
32. Maximum Slenderness Ratio for members normally in pure tension: **400**.
33. Partial Safety Factor for yield of structural steel (γ_{m0}): **1.10**.
34. Partial Safety Factor for shop-welded connections (γ_{mw}): **1.25**.
35. Partial Safety Factor for field-welded connections (γ_{mw}): **1.50**.
36. Deflection Limit for gantry girders with manually operated cranes: **Span / 500**.
37. Deflection Limit for gantry girders with electrically operated cranes: **Span / 750**.
38. Permissible Bending Stress in steel base plates under WSM (IS 800-1984): **1500 kg/cm²**.
39. Standard Shape Factor for a rectangular section: **1.50**.
40. Standard Shape Factor for a solid circular section: **1.70**.
41. Standard Shape Factor for a diamond section (diagonal horizontal): **2.00**.
42. Standard Shape Factor for a thin-walled hollow circular section: **1.27**.
43. Standard Shape Factor for an isosceles triangular section: **2.34**.
44. Average Shape Factor of standard symmetrical I-beams (major axis): **1.14**.
45. Average Shape Factor of standard symmetrical I-beams (minor axis): **1.50**.

46. Average Shape Factor of asymmetrical Tee-sections: **1.75 to 1.80**.
47. Local slenderness of column component between lacing bars: ≤ 50 or $\leq 0.7 \cdot \lambda_{\text{whole}}$.
48. Minimum thickness of gusset plate for standard bases: **12 mm**.
49. Minimum cover of concrete encasement for grillage foundations: **100 mm**.
50. Standard live load on sloping roof without access (slope $\leq 10^\circ$): **0.75 kN/m²**.

PART 4: TOP 50 RECURRING EXAMINER TRAPS

1. Sizing the footprint area of base plates or grillage bases using factored ultimate loads instead of working serviceability loads.
2. Using the nominal bolt/rivet diameter d in shearing and bearing calculations instead of the gross hole diameter d_g .
3. Forgetting to divide the total lacing transverse shear force (2.5% of column axial load) by the number of parallel lacing planes.
4. Applying lacing transverse shear design load checks without incorporating lacing inclination angle trigonometric multipliers.
5. Ignoring the 10% slenderness penalty (effective slenderness ratio $= 1.10 \cdot \lambda$) in intermediate batten column design checks.
6. Forgetting to subtract countersunk depth or applying a 33.3% capacity reduction factor for countersunk rivets in tension.
7. Assuming that high-strength steel grades increase the shape factor of a cross-section (shape factor is independent of steel grade).
8. Substituting overall beam depth D instead of effective depth d in rolled beam shear strength capacity checks.
9. Applying simply supported beam elastic deflection formulas to plastic collapse mechanisms where hinges alter kinematics.
10. Confusing the maximum pitch limit of fasteners in tension (16t) with the maximum pitch limit in compression (12t).
11. Using the larger radius of gyration (r_{xx}) instead of weak-axis minimum (r_{yy}) to check slenderness ratio of standard channel columns.
12. Assuming the elastic neutral axis (centroid) and plastic neutral axis (equal-area axis) coincide for asymmetrical sections like Tees.
13. Assuming the critical fastener in an eccentric joint is the furthest one, without verifying the angle between force vectors.
14. Designing welds for combined normal and shear stresses by direct arithmetic summation instead of using equivalent stress limits.
15. Confusing the theoretical column effective length factors (e.g., 0.50L) with the conservative design recommended ones (e.g., 0.65L).
16. Using gross area instead of net area when checking the design tensile strength of plates under tension in limit state design.
17. Swapping the partial safety factor for steel yield ($\gamma_{m0} = 1.10$) with the ultimate strength factor ($\gamma_{m1} = 1.25$).
18. Failing to divide the concrete bearing load by 2 when column bases are stiffened with double-tier perpendicular grillage beam tiers.
19. Substituting plate width instead of plate thickness in maximum fillet weld square edge limits (size $\leq t - 1.5$ mm).
20. Evaluating propped cantilever collapse loads under central load as $8 \cdot M_p / L$ instead of the correct $6 \cdot M_p / L$.
21. Using the standard fusion constant $k = 0.70$ for non-90° fillet weld angles where fusion face angles reduce k to 0.50.
22. Assuming the shape factor of hollow circular tubes is identical to solid circular sections (1.27 vs. 1.70).
23. Omitting the self-weight of grillage beams and concrete encasement (typically taken as 10% of axial load) when sizing the footing.
24. Swapping the Web Crippling formula ($b_1 + n_2$) with the Web Buckling formula ($b_1 + n_1$) dispersion lengths.
25. Swapping the lacing slenderness multiplier (1.05) with the batten slenderness multiplier (1.10).
26. Calculating the slenderness ratio of a lacing bar using total column length instead of lacing bar length between connections.
27. Ignoring the 0.02 kN/m² live load reduction per degree for roof slopes greater than 10° up to 15°.
28. Assuming the shear stress distribution in an I-beam is uniform across the entire depth of the section.
29. Confusing the elastic section modulus (Z_e) with the plastic section modulus (Z_p) when calculating bending stresses.
30. Forgetting that weld strengths are calculated on the throat thickness and not on the weld size (s).
31. Assuming that support settlements or thermal movements reduce the ultimate plastic collapse load of a beam.
32. Stating that the lower bound theorem of plastic analysis yields an unsafe (upper bound) estimate of collapse load.
33. Stating that the upper bound theorem of plastic analysis yields a safe (lower bound) estimate of collapse load.

34. Designing gusseted column bases without including the concrete pedestal bearing pressure check.
35. Calculating the volume of a weld using the nominal plate thickness instead of the weld throat area.
36. Swapping the pitch of bolts along the direction of stress with the gauge of bolts perpendicular to stress.
37. Using the nominal bolt diameter instead of gross hole diameter for calculating gross block shear areas.
38. Assuming that the thickness of the column web has no effect on the required thickness of the column slab base plate.
39. Ignoring local buckling of flange cover plates in built-up beams by exceeding the $12 \cdot t$ compression pitch limit.
40. Swapping the functions of gantry girder crane rail support elements: channel attached to top flange resists lateral forces.
41. Assuming that the King Post in a roof truss is a compression member (it is strictly a vertical tension tie).
42. Forgetting that the effective length of a column is doubled ($2L$) when one end is fixed and the other is free.
43. Substituting the wrong radius of gyration (r_{min}) for unequal angle discontinuous struts (must use r_w).
44. Assuming that the plastic neutral axis of an unsymmetrical Tee section lies at the centroidal axis.
45. Calculating equivalent stresses in fillet welds without multiplying the longitudinal shear stress by the square root of 3.
46. Checking lateral-torsional buckling of beams using the gross area instead of the compression flange area.
47. Designing batten plates under transverse shear without validating double bending and shear checks at intermediate joints.
48. Substituting the nominal diameter instead of the gross bolt diameter in block shear strength calculations.
49. Assuming the live load on a flat roof or roof with a slope $\leq 10^\circ$ is 1.50 kN/m^2 instead of the correct 0.75 kN/m^2 .
50. Forgetting that bending an I-section about its minor axis (y-y) raises its shape factor to 1.50.

PART 5: TOP 25 "NEVER FORGET" CONCEPTS

1. Young's Modulus of structural steel is a metallurgical constant (200 GPa) identical for all grades from Fe250 to Fe540.
2. Under IS 2062, structural steel has a maximum carbon content of 0.23% to prevent weld zone embrittlement and preserve ductility.
3. In rolled I-beams, the flanges resist major bending moments, while the web resists over 95% of the total transverse shear force.
4. Web buckling is a column-like buckling failure of the web, whereas web crippling is local material crushing at load points.
5. Unwin's formula $d = 6.01 \cdot \sqrt{t}$ determines the nominal rivet diameter for plate thicknesses $t \geq 8 \text{ mm}$.
6. The gross diameter of a rivet hole (d_g) is $d + 1.5 \text{ mm}$ for $d \leq 25 \text{ mm}$, and $d + 2.0 \text{ mm}$ for $d > 25 \text{ mm}$.
7. Joint efficiency (η) is based strictly on the minimum of shearing, bearing, and tearing capacities divided by solid plate strength.
8. The effective throat thickness of a fillet weld is $t_e = k \cdot s$, where s is the weld size and k is the fusion constant.
9. Combined stresses in fillet welds are checked using the equivalent stress criterion: $f_e = \sqrt{[f_a^2 + 3 \cdot q^2]} \leq f_y / \gamma_{m0}$.
10. Unsymmetric weld groups must balance weld lengths to align the weld centroid with the gravity axis of the connected member.
11. The critical bolt/rivet in an eccentric connection is the one that has the maximum vector sum of direct and twisting shear stresses.
12. Column slenderness ratio is $\lambda = L_e / r_{min}$, and columns buckling occurs about the weakest axis with minimum radius of gyration.
13. Built-up laced columns have slenderness ratio increased by 5%, and battened columns by 10% to account for shear deformation.
14. Lacing systems are designed as pin-jointed trusses to resist a transverse shear force equal to 2.5% of the column's axial load.
15. Batten plates are designed as rigid frames subjected to combined bending and shear due to column's transverse shear force.
16. Slab bases are flat steel plates, whereas gusseted bases use gusset plates and angles to stiffen the base plate and reduce thickness.
17. The maximum permissible bending stress in steel slab bases is limited to 1500 kg/cm^2 (150 MPa) under WSM.
18. Grillage foundations distribution checks are based strictly on serviceability working loads, never factored ultimate loads.
19. Purlins are continuous horizontal beams spanning between roof trusses, designed for combined axial and bending stresses.
20. Plastic moment capacity is $M_p = Z_p \cdot f_y$, representing the maximum moment a section can carry when fully plastic.

21. In plastic bending, the plastic neutral axis (PNA) divides the cross-section into two equal areas ($A_c = A_t = A/2$).
22. The shape factor (S) is the ratio of plastic to elastic section modulus (Z_p / Z_e), representing structural reserve strength.
23. Symmetrical rolled steel I-sections have major axis shape factor of 1.12 - 1.15, which increases to 1.50 under minor axis bending.
24. The lower bound theorem of plastic analysis yields a safe estimate, whereas the upper bound theorem yields an unsafe estimate.
25. Propped cantilever collapse load under central load is $6 \cdot M_p / L$, and under UDL is $11.656 \cdot M_p / L^2$.

APTRANSCO CIVIL AEE HIGH-YIELD REVISION GUIDE

SOIL MECHANICS FINAL EXAM KIT • SYLLABUS CORE REVISION

Technical Context & Purpose: This compendium represents the exhaustive, high-density revision notes for the Soil Mechanics portion of the APTRANSCO Civil Assistant Executive Engineer (AEE) syllabus. Mapped strictly to chapters covering physical properties, classifications, field identifications, permeability, capillarity, seepage flow nets, compaction control, Mohr-Coulomb shear strength, lateral earth pressures, and slope stability. All mathematical design formulations are typeset in proper technical notation, and double-bullet formatting errors are strictly eliminated. All advanced deep bearing capacities, pile caps, and foundation settlement analyses have been excluded from this booklet as they belong to the separate Foundation Engineering curriculum.

PART 1: TOP 60 GOLDEN FACTS (CORE CONCEPT RECONNAISSANCE)

1. Soil is a heterogeneous multi-phase engineering system consisting of solid mineral grains, pore water, and pore air.
2. Residual soils are formed by the in-situ chemical or mechanical weathering of rocks and remain at their site of origin.
3. Alluvial soils are transported and deposited by the physical action of running water in rivers, channels, and floodplains.
4. Lacustrine soils are deposited in quiet, static, fresh-water lakes, resulting in highly laminated varved clay deposits.
5. Loess is a wind-borne, aeolian silt deposit characterized by high void ratios, uniform grading, and collapsible structural stability.
6. Colluvial soil (Talus) is transported by gravitational mass wasting and deposited in random piles at the base of steep cliffs.
7. Bentonite is a highly plastic, sodium-rich clay formed by the chemical weathering of volcanic ash deposits.
8. Black cotton soil is a residual clay rich in montmorillonite mineral, derived from the weathering of basalt or trap rocks.
9. Kaolinite possesses a 1:1 crystal sheet structure held by strong hydrogen bonds, resulting in highly stable, non-swelling clays.
10. Illite possesses a 2:1 crystal sheet structure with potassium (K^+) ions providing a stable interlayer ionic bond.
11. Montmorillonite has a 2:1 crystal sheet structure with weak Van der Waals bonds, causing extreme swelling and shrinkage.
12. Soil texture describes the relative percentage of different grain size fractions and is a permanent physical feature.
13. Soil structure (fabric) describes the geometric arrangement of soil particles, which is highly susceptible to mechanical disturbance.
14. Coarse-grained soils exhibit single-grained or honeycomb structures dominated strictly by gravitational forces.
15. Fine-grained clay particles form flocculated (edge-to-face) or dispersed (face-to-face) structures under electrostatic forces.
16. Flocculated clay fabric exhibits high shear strength, high hydraulic conductivity, and low compressibility under loading.
17. Dispersed clay fabric exhibits low shear strength, low hydraulic conductivity, and high compressibility under surcharge.
18. Void ratio (e) is defined as the volumetric ratio of the volume of voids (V_v) to the volume of solid soil grains (V_s).
19. Porosity (n) is defined as the volumetric ratio of the volume of voids (V_v) to the total volume of the soil mass (V).
20. Void ratio (e) has no physical upper limit and can exceed 1.0, but porosity (n) must always remain strictly less than 100%.

21. Degree of saturation (S_r) represents the volumetric ratio of the volume of water to the total volume of void spaces.
22. Air content (a_c) is the ratio of the volume of air to the volume of voids, acting as the mathematical complement to saturation.
23. Saturated unit weight represents the bulk unit weight of soil when all voids are completely filled with water ($S_r = 1.0$).
24. Submerged unit weight is the effective unit weight of soil under buoyancy, which is approximately equal to half of the saturated weight.
25. Specific gravity of soil solids is the ratio of the unit weight of solids to the unit weight of standard water.
26. Specific gravity of organic soils (1.2 to 1.8) is significantly lower than that of inorganic mineral soils (2.6 to 2.8).
27. Stokes' Law governs the terminal sedimentation velocity of spherical particles falling freely through a viscous fluid medium.
28. Mechanical sieve analysis is suitable for coarse soils, whereas hydrometer and pipette sedimentation tests are used for fine soils.
29. Well-graded soils contain a wide and even distribution of grain sizes, resulting in high compacted density and low voids.
30. Uniformly graded soils contain particles of nearly identical sizes, resulting in high void ratios and low compacted densities.
31. Gap-graded soils are characterized by the complete absence of one or more intermediate particle size fractions.
32. Atterberg limits define the boundaries of water content at which a fine soil transitions between different consistency states.
33. Liquid limit (w_L) is the water content at which soil transitions from a plastic state to a liquid state with negligible shear strength.
34. Plastic limit (w_p) is the water content at which soil can be rolled into a 3 mm thread without showing signs of crumbling.
35. Shrinkage limit (w_s) is the maximum water content at which further moisture loss causes absolutely no volume reduction.
36. Plasticity index (I_p) is the numerical difference between the liquid limit and the plastic limit of a fine-grained soil.
37. Activity of clay represents the ratio of the plasticity index to the percentage of clay-sized particles smaller than 2 microns.
38. Darcy's Law states that flow velocity is directly proportional to the hydraulic gradient under strictly laminar flow regimes.
39. Constant head permeameter is suitable for coarse sands, whereas falling head permeameter is used for fine-grained silts and clays.
40. Seepage velocity (v_s) is always greater than discharge velocity (v) because flow is restricted to the pore void channels.
41. Capillary rise in soils is caused by surface tension and is inversely proportional to the effective pore diameter.
42. Capillary suction creates negative pore water pressure, providing temporary apparent cohesion in silty soils and fine sands.
43. Total stress is the sum of the effective stress carried by the soil skeleton and the pore water pressure.
44. Effective stress controls all primary engineering properties of soil, including its shear strength and volumetric compressibility.
45. A rise in the groundwater table above the ground surface causes absolutely no change in the effective stress at any depth.
46. Seepage force is the viscous drag force exerted by flowing water on the soil particles in the direction of seepage.

47. Upward seepage decreases the effective stress of the soil, whereas downward seepage increases the effective stress.
48. Quicksand condition occurs when upward seepage pressure completely neutralizes the effective overburden stress of cohesionless sand.
49. Critical hydraulic gradient (i_c) represents the gradient at which the quicksand condition initiates in sand deposits.
50. A flow net is a graphical representation of 2D steady-state seepage consisting of orthogonal flow lines and equipotential lines.
51. The quantity of seepage through a flow net is proportional to the ratio of flow channels to equipotential drops.
52. Exit gradient is the hydraulic gradient at the downstream face where seepage water exits the soil mass.
53. Piping failure initiates at the downstream toe when the exit gradient exceeds the critical hydraulic gradient of the sand.
54. Soil compaction is the mechanical expulsion of pore air, resulting in a rapid increase in dry density.
55. Compaction is an instantaneous process, whereas consolidation is a time-dependent expulsion of pore water from saturated clays.
56. Sheepsfoot rollers compact soil from the bottom up using kneading action, making them ideal for cohesive clays.
57. Vibratory rollers are highly effective for compacting cohesionless sands by utilizing dynamic resonance frequency.
58. Dry of optimum compaction yields a flocculated soil structure, whereas wet of optimum yields a dispersed structure.
59. The Zero Air Voids line represents the theoretical maximum dry density of soil at 100% saturation (zero air content).
60. Clay soils can never be compacted past the Zero Air Voids line under any compactive effort on site.

PART 2: TOP 50 GOVERNING FORMULAS (GEOTECHNICAL EQUATION RIGOR)

1. Void Ratio: $e = V_v / V_s$
2. Porosity: $n = V_v / V$
3. Porosity to Void Ratio: $e = n / (1 - n)$
4. Void Ratio to Porosity: $n = e / (1 + e)$
5. Degree of Saturation: $S_r = V_w / V_v$
6. Air Content: $a_c = V_a / V_v = 1 - S_r$
7. Percentage Air Voids: $n_a = V_a / V = n \cdot a_c$
8. Water Content: $w = W_w / W_s$
9. Bulk Unit Weight: $\gamma = G_s \cdot \gamma_w \cdot (1 + w) / (1 + e)$
10. Saturated Unit Weight: $\gamma_{sat} = \gamma_w \cdot (G_s + e) / (1 + e)$
11. Dry Unit Weight (Phase): $\gamma_d = G_s \cdot \gamma_w / (1 + e)$
12. Dry Unit Weight (Bulk): $\gamma_d = \gamma / (1 + w)$
13. Submerged Unit Weight: $\gamma' = \gamma_{sat} - \gamma_w = \gamma_w \cdot (G_s - 1) / (1 + e)$
14. Saturated-Void-Water Relation: $S_r \cdot e = w \cdot G_s$
15. Dry Unit Weight with Air Voids: $\gamma_d = (1 - n_a) \cdot G_s \cdot \gamma_w / (1 + w \cdot G_s)$
16. Coefficient of Uniformity: $C_u = D_{60} / D_{10}$
17. Coefficient of Curvature: $C_c = D_{30}^2 / (D_{60} \cdot D_{10})$
18. Density Index (Relative Density): $I_D = (e_{max} - e) / (e_{max} - e_{min})$
19. Plasticity Index: $I_p = w_L - w_P$
20. Liquidity Index: $I_L = (w - w_P) / I_p$
21. Consistency Index: $I_C = (w_L - w) / I_p$
22. Flow Index (Liquid Limit): $I_f = (w_1 - w_2) / \log_{10}(N_2 / N_1)$
23. Toughness Index: $I_t = I_p / I_f$
24. Skempton's Clay Activity: $A = I_p / (\% \text{ Clay Fines})$
25. Plasticity Chart A-Line: $I_p = 0.73 \cdot (w_L - 20)$
26. Plasticity Chart U-Line: $I_p = 0.90 \cdot (w_L - 8)$
27. Darcy's Law: $v = k \cdot i$
28. Seepage Velocity: $v_s = v / n$
29. Constant Head Permeability: $k = Q \cdot L / (A \cdot h \cdot t)$
30. Falling Head Permeability: $k = 2.303 \cdot (a \cdot L / (A \cdot t)) \cdot \log_{10}(h_1 / h_2)$
31. Equivalent Horizontal Permeability: $k_h = \Sigma(k_i \cdot H_i) / \Sigma H_i$
32. Equivalent Vertical Permeability: $k_v = \Sigma H_i / \Sigma(H_i / k_i)$
33. Capillary Rise: $h_c = 4 \cdot \sigma \cdot \cos(\theta) / (\rho_w \cdot g \cdot d)$
34. Empirical Capillary Rise: $h_c \approx 0.3 / d$ (in cm)
35. Effective Stress (Static): $\sigma' = \sigma - u$
36. Seepage Force: $F_s = i \cdot \gamma_w \cdot V$
37. Critical Hydraulic Gradient: $i_c = (G_s - 1) / (1 + e) = \gamma' / \gamma_w$
38. Quantity of Seepage (Flow Net): $q = k \cdot H \cdot (N_f / N_d)$

39. Exit Gradient (Flow Net): $i_{exit} = \Delta h / l$
40. Mohr-Coulomb Shear Strength: $\tau = c + \sigma \cdot \tan(\phi)$
41. Effective Mohr-Coulomb Strength: $\tau' = c' + \sigma' \cdot \tan(\phi')$
42. Failure Plane Angle: $\theta_f = 45^\circ + \phi/2$
43. Principal Stress Relation at Failure: $\sigma_1 = \sigma_3 \cdot \tan^2(45^\circ + \phi/2) + 2 \cdot c \cdot \tan(45^\circ + \phi/2)$
44. Unconfined Compression Strength: $q_u = 2 \cdot c_u$
45. Rankine Active Earth Pressure Coefficient: $K_a = (1 - \sin(\phi)) / (1 + \sin(\phi))$
46. Rankine Passive Earth Pressure Coefficient: $K_p = (1 + \sin(\phi)) / (1 - \sin(\phi))$
47. Rankine Earth Pressure Coefficient At-Rest: $K_0 = 1 - \sin(\phi)$
48. Unsupported Cut Depth: $H_c = 4 \cdot c / (\gamma \cdot \sqrt{K_a})$
49. Infinite Slope Safety Factor (Cohesionless): $F_s = \tan(\phi) / \tan(\beta)$
50. Taylor's Stability Number: $S_n = c / (F_s \cdot \gamma \cdot H)$

PART 3: TOP 50 STANDARD VALUES & CODAL LIMITS (STATUTORY REFERENCE)

1. Unit weight of standard water: **9.81 kN/m³** (or $9.81 \times 10^3 \text{ N/m}^3$).
2. Mass density of standard water: **1000 kg/m³** (or 1.0 g/cm³).
3. Oven drying temperature limit for standard inorganic mineral soils: **105°C to 110°C**.
4. Oven drying temperature limit for organic soils or gypsum-rich soils: **60°C to 80°C** to prevent gypsum decomposition.
5. Specific gravity range for standard quartz-based sand particles: **2.63 to 2.67**.
6. Specific gravity range for inorganic clay minerals: **2.70 to 2.80**.
7. IS Sieve size dividing coarse-grained and fine-grained soils: **75 microns** (0.075 mm).
8. IS Sieve size dividing gravel and sand fractions: **4.75 mm**.
9. IS Sieve size dividing coarse sand and medium sand: **2.00 mm**.
10. IS Sieve size dividing medium sand and fine sand: **425 microns** (0.425 mm).
11. Particle size limit below which Stokes' Law remains valid: **0.2 mm**.
12. Particle size limit below which Brownian movement disrupts Stokes' Law: **0.0002 mm** (0.2 microns).
13. Minimum Coefficient of Uniformity for well-graded gravel: **$C_u > 4$** .
14. Minimum Coefficient of Uniformity for well-graded sand: **$C_u > 6$** .
15. Range of Coefficient of Curvature for well-graded soils: **$1.0 \leq C_c \leq 3.0$** .
16. Clay content percentage threshold particle size used in Skempton's activity: **< 2 microns**.
17. Standard number of drops per second in Casagrande's Liquid Limit device: **2 drops/second**.
18. Standard height of soil pat drop in Casagrande's Liquid Limit device: **10 mm** ($\pm 0.1 \text{ mm}$).
19. Standard number of blows required to define the Liquid Limit: **25 blows**.
20. Thread diameter used to define the Plastic Limit: **3 mm**.
21. Minimum plasticity index of clay as per Casagrande's chart: **$I_p > 7\%$** .
22. Plasticity index range for dual symbol classification (e.g., CL-ML): **4% to 7%**.
23. Liquid limit dividing low and medium compressibility inorganic soils: **35%**.
24. Liquid limit dividing medium and high compressibility inorganic soils: **50%**.
25. Permeability coefficient range for clean gravels: **$k > 1.0 \text{ cm/sec}$** .
26. Permeability coefficient range for clean sands: **$k = 1.0 \text{ to } 10^{-3} \text{ cm/sec}$** .
27. Permeability coefficient range for silts and silty sands: **$k = 10^{-3} \text{ to } 10^{-7} \text{ cm/sec}$** .
28. Permeability coefficient range for homogeneous clays: **$k < 10^{-7} \text{ cm/sec}$** .
29. Critical hydraulic gradient value for typical natural sand deposits: **$i_c \approx 1.0$** .
30. Surface tension of standard water at 20°C against air: **0.073 N/m**.
31. Standard compaction mold height for Proctor tests: **127.3 mm** (5.0 inches).
32. Standard mold volume for Standard Proctor Test: **1000 cm³** (or 1/30 of a cubic foot).
33. Weight of rammer for Standard Proctor Test: **2.6 kg** (or 5.5 lbs under ASTM / 2.49 kg under IS).
34. Drop height of rammer for Standard Proctor Test: **310 mm** (or 12 inches).
35. Number of soil layers in Standard Proctor Test: **3 layers**.
36. Number of blows per layer in Standard Proctor Test: **25 blows**.
37. Weight of rammer for Modified Proctor Test: **4.9 kg** (or 10.0 lbs under ASTM / 4.89 kg under IS).
38. Drop height of rammer for Modified Proctor Test: **450 mm** (or 18 inches).

- 39. Number of soil layers in Modified Proctor Test: **5 layers**.
- 40. Energy ratio of Modified Proctor Test to Standard Proctor Test: **4.55 times**.
- 41. Proctor needle penetration speed in compaction control: **1.25 cm/second**.
- 42. Standard dimensions of core cutter cylinder: **100 mm** diameter, **130 mm** height.
- 43. Soil failure plane angle for saturated clay under UCC test ($\phi = 0$): $\theta_f = 45^\circ$.
- 44. Pore pressure coefficient B value for fully saturated soils: **1.0**.
- 45. Pore pressure coefficient B value for completely dry soils: **0**.
- 46. Limit of active pressure coefficient K_a for $\phi = 30^\circ$: **0.33** (or $1/3$).
- 47. Limit of passive pressure coefficient K_p for $\phi = 30^\circ$: **3.0**.
- 48. Critical depth-to-width ratio dividing shallow and deep foundations: $D_f / B \leq 1$.
- 49. Standard factor of safety against sliding in slope stability: **1.50**.
- 50. Standard factor of safety against overturning in slope stability: **2.00**.

PART 4: TOP 50 RECURRING EXAMINER TRAPS (TACTICAL WARNING DIRECTORY)

1. Swapping void ratio (e) and porosity (n) formulas, forgetting that total volume V includes solid volume V_s .
2. Entering water content as a whole percentage in the $Se = wG$ formula instead of converting it to a decimal.
3. Calculating dry unit weight as $\gamma \cdot (1 - w)$ instead of utilizing the correct reciprocal form $\gamma / (1 + w)$.
4. Assuming porosity can exceed 100% in highly porous soils, forgetting solid volume is always non-zero.
5. Confusing the specific gravity of standard soil solids with mass specific gravity of soil.
6. Forgetting to apply temperature correction factors when specific gravity is measured at non-standard temperatures.
7. Drying organic soils at 110°C, which destroys organic matter and yields an incorrect water content.
8. Omitting the 40% blow energy correction when using a larger non-standard compaction mold volume.
9. Rounding intermediate void ratio decimals too early in multi-phase soil calculations, causing massive cumulative errors.
10. Assuming well-graded soils require only a high C_u , neglecting the C_c limit range of 1 to 3.
11. Using the total specimen height instead of the drainage path $d = H/2$ under double drainage.
12. Confusing the standard rammer drop heights of Standard Proctor (12 inches) with Modified Proctor (18 inches).
13. Swapping the active and passive earth pressure coefficients, forgetting that K_a is always less than K_p .
14. Using bulk unit weight instead of submerged unit weight for soil below the groundwater table.
15. Substituting nominal core cutter diameter instead of effective internal radius in volume calculations.
16. Assuming capillary rise is higher in coarse sand than in fine silt due to larger flow channels.
17. Believing that a rise in the water table above the ground surface increases the effective stress at depth.
18. Swapping the critical sections of footings: evaluating punching shear at d instead of $d/2$.
19. Using a linear stress-depth relationship below a point load, ignoring Boussinesq's inverse-square law.
20. Forgetting that the flow net seepage quantity formula uses the head loss H , not total water depth.
21. Confusing the critical hydraulic gradient with the exit gradient in piping failure calculations.
22. Assuming sheepfoot rollers compact dry sand effectively, when they are strictly designed for cohesive clays.
23. Swapping the liquid limit dividing lines: using 50% for low compressibility instead of 35%.
24. Using the overall thickness instead of the drainage path distance in Thiem's unconfined well equations.
25. Neglecting the effect of seepage forces in infinite slope stability, overestimating the safety factor.
26. Confusing the flow index I_f with the plasticity index I_p in toughness index calculations.
27. Substituting percentage air voids n_a instead of air content a_c in three-phase weight relations.
28. Assuming the failure plane in a Direct Shear Test is the weakest plane, when it is actually predetermined.
29. Omitting the pore pressure correction when calculating effective stress under upward seepage.
30. Using Terzaghi's shallow bearing factor $N_c = 5.7$ for driven piles, which strictly require $N_c = 9.0$.
31. Assuming relative compaction and relative density are identical, ignoring Lee's empirical offset.
32. Forgetting that the unconfined compression strength q_u is twice the undrained cohesion c_u .
33. Using the nominal diameter of a soil sampler instead of the inner cutting edge diameter to compute area ratios.
34. Assuming that Coulomb's wedge theory neglects wall friction, which is actually its key advantage over Rankine.
35. Swapping the coefficient of compressibility a_v with the coefficient of volume compressibility m_v .
36. Forgetting to convert angles from degrees to radians in the strip load vertical stress formula.
37. Using the wet unit weight in Taylor's stability number under fully submerged conditions.

38. Assuming that the A-line separates clays and silts at all liquid limits, ignoring the ML-CL transition zone.
39. Swapping the definitions of thixotropy (isothermal strength recovery) and sensitivity (strength loss on remolding).
40. Confusing constant head tests with falling head tests, attempting to use constant head on fine-grained clays.
41. Forgetting that the critical depth of skin friction in piles is capped at 15 to 20 times the pile diameter.
42. Assuming that the active zone of black cotton clay is stable, when it is the highly unstable swell-shrink zone.
43. Confusing the major principal stress with the minor principal stress in active earth pressure states.
44. Using gross area instead of net concrete area in structural column capacity checks.
45. Forgetting that the negative skin friction in piles requires a safety factor of 3.0 instead of 2.5.
46. Swapping the hydraulic gradient coefficient R_{w1} and R_{w2} under intermediate water table levels.
47. Assuming that the tension crack depth is the maximum depth of unsupported excavation, which is actually twice that depth.
48. Neglecting the self-weight of the soil block when calculating pile group capacity in block mode.
49. Using Thiem's unconfined discharge formula for confined aquifers, leading to massive flow estimation errors.
50. Swapping the vertical and horizontal equivalent permeabilities, forgetting that parallel flow k_h is always greater than series flow k_v .

PART 5: TOP 25 'NEVER FORGET' CONCEPTS (NON-NEGOTIABLE RECONNAISSANCE)

1. Saturated clays undergoing unconfined compression tests fail strictly along a plane oriented at 45° .
2. Soil compaction expels pore air, whereas consolidation expels pore water over time.
3. Capillary rise is inversely proportional to particle size, peaking in fine-grained silts and clays.
4. Effective stress controls the shear strength and volumetric change behavior of all soil skeletons.
5. Zero Air Voids density is a theoretical upper limit that can never be exceeded under any compactive effort.
6. Well-graded gravels require a Coefficient of Uniformity greater than 4 and a Coefficient of Curvature between 1 and 3.
7. Well-graded sands require a Coefficient of Uniformity greater than 6 and a Coefficient of Curvature between 1 and 3.
8. The critical hydraulic gradient represents a complete loss of shear strength in cohesionless soils under upward seepage.
9. Clay soils compacted dry of optimum exhibit high strength and high permeability due to flocculated fabric.
10. Clay soils compacted wet of optimum exhibit low strength and low permeability due to dispersed fabric.
11. Apparent cohesion in dry sand is zero, but slight moisture creates capillary meniscus tension which disappears on inundation.
12. Flow nets can only be drawn for steady-state, laminar, and irrotational flow regimes.
13. Exit gradient must be kept well below the critical hydraulic gradient to prevent downstream piping failure.
14. Unconfined compressive strength is strictly equal to twice the undrained cohesive strength for clay.
15. Passive earth pressure requires significantly larger lateral strain to fully mobilize compared to active earth pressure.
16. The net ultimate bearing capacity of purely cohesive clay under Terzaghi's theory is strictly independent of footing size.
17. Skempton's pore pressure coefficient B is equal to 1.0 only when the soil is 100% saturated.
18. Active earth pressure is the minimum lateral pressure, whereas passive earth pressure is the maximum lateral pressure.
19. Taylor's stability number is a dimensionless parameter used to analyze the stability of finite slopes.
20. Coulomb's earth pressure theory accounts for wall friction and inclined backfaces, whereas Rankine's theory assumes smooth vertical walls.
21. Sensitivity represents the loss of soil strength upon remolding at constant water content.
22. Thixotropy is the isothermal, time-dependent recovery of shear strength lost during remolding.
23. Specific gravity of organic soils is lower than inorganic soils due to high carbonaceous content.
24. Relative compaction measures the field dry density relative to the laboratory maximum dry density.
25. A rise in the water table above the ground surface has no effect on the effective stress at depth under hydrostatic conditions.

APTRANSCO AEE (CIVIL) FINAL EXAM KIT

FOUNDATION ENGINEERING (Chapters 1 to 12) — Last 30-Minute Rapid Revision

PART 1 — TOP 60 GOLDEN FACTS

1. Boussinesq's point load theory assumes a semi-infinite, elastic, homogeneous, isotropic, and weightless soil medium.
2. Westergaard's theory models anisotropic layered soils reinforced by thin, inextensible horizontal sheets.
3. Westergaard's vertical stress under a concentrated point load is smaller than Boussinesq's directly along the load line.
4. Boussinesq and Westergaard stress distribution curves intersect at a radial-to-depth ratio r/z of exactly 1.52.
5. Newmark's influence value is defined as the reciprocal of the total area units: $I_v = 1 / (\text{Number of rings} \cdot \text{Number of radial lines})$.
6. On a Newmark influence chart, the scale bar length AB is drawn exactly equal to the target calculation depth z .
7. If target calculation depth z is doubled, the footprint of the footing plotted on a Newmark chart must be halved.
8. The maximum horizontal width of a point load pressure bulb occurs at a depth of exactly 0.775 times its maximum depth.
9. Relative isobars remain constant under load doubling, whereas absolute isobars scale by the square root of 2.
10. In sand, contact pressure is maximum at the center and zero at the edges under a completely rigid footing.
11. In clay, contact pressure is minimum at the center and theoretically infinite at the edges under a completely rigid footing.
12. Flexible footings on any soil produce a completely uniform contact pressure profile equal to the applied load.
13. Rigid footings on cohesive soil produce a completely uniform settlement profile while contact pressure varies.
14. Footing self-weight is strictly omitted during bending moment and shear capacity calculations of footing slabs.
15. Soil reactions represent the upward pressure of the ground resisting downward structural foundation loads.
16. Terzaghi's bearing capacity theory assumes general shear failure, rough base, shallow depth, and continuous strip footprint.
17. The depth of a shallow foundation must be less than or equal to its least lateral dimension (depth-to-width ratio ≤ 1.0).
18. General shear failure occurs suddenly with a distinct failure surface extending to the ground level and sudden tilting.
19. Local shear failure occurs in medium-dense soils with progressive plastic yielding and no sudden tilting or clear surface slip.
20. Punching shear failure is characterized by vertical soil shearing around the footing perimeter with zero surface heave.
21. Local shear failure requires reduced design parameters: $c' = (2/3) \cdot c$ and $\tan(\phi') = (2/3) \cdot \tan(\phi)$.

22. Terzaghi's bearing capacity factors N_c , N_q , N_γ are dimensionless constants depending solely on the internal friction angle ϕ .
23. For purely cohesive soils ($\phi = 0$), Terzaghi's shallow strip factors are strictly $N_c = 5.7$, $N_q = 1.0$, and $N_\gamma = 0$.
24. Groundwater table rise above the base of a footing reduces the effective surcharge and soil unit weight terms.
25. When the water table is at or above the ground surface, the soil submerged unit weight γ' must be used in all capacity terms.
26. Plate load test results scale directly by width for sands, but are independent of plate size for purely cohesive clays.
27. Net safe bearing capacity is defined as net ultimate bearing capacity divided by the safety factor ($q_{ns} = q_{nu} / F$).
28. Gross safe bearing capacity adds the stabilizing overburden back: $q_s = (q_u - q) / F + q$.
29. Permissible total settlement limit as per IS:1904 is 65 mm for isolated footings on clay and 40 mm on sand.
30. Permissible differential settlement limit under IS:1904 is 40 mm for isolated pads on clay and 25 mm on sand.
31. Terzaghi's 1D consolidation theory assumes homogeneous, saturated soil with purely vertical one-way water drainage.
32. Under double drainage conditions, the maximum water travel path d is exactly half of the total clay layer thickness H .
33. Time required to reach a specific degree of consolidation scales with the square of the drainage path ($t \propto d^2$).
34. The coefficient of compressibility a_v is the slope of the void ratio versus effective stress curve on arithmetic axes.
35. The coefficient of volume compressibility m_v represents the volumetric strain per unit stress increase: $m_v = a_v / (1 + e_0)$.
36. Compression index C_c represents the slope of the virgin compression line plotted on a semi-log stress scale.
37. Preconsolidation pressure is the maximum effective vertical stress that the soil has ever experienced in its geological history.
38. Overconsolidation ratio (OCR) is defined as preconsolidation stress divided by the current in-situ effective stress.
39. Normally consolidated clay has an OCR of exactly 1.0, whereas overconsolidated clay has an OCR greater than 1.0.
40. Skempton's pore pressure coefficient B is exactly 1.0 for completely saturated soils, and zero for dry soils.
41. Pile foundations are categorized as end-bearing, friction, compaction, batter, or fender piles based on load function.
42. End-bearing capacity of piles in purely cohesive clay utilizes a deep bearing factor N_c equal to exactly 9.0.
43. Friction resistance of piles in clay is evaluated as $Q_s = \alpha \cdot c_u \cdot A_s$, where α is the empirical adhesion factor.
44. Adhesion factor α is close to 1.0 for soft clays, but drops to 0.3 or 0.4 for stiff, overconsolidated clays.
45. The Engineering News Formula utilizes a built-in safety factor F of 6.0 for predicting dynamic pile capacity.
46. Safe dynamic load constants in the ENR formula are $C = 2.54$ cm for drop hammers and 0.254 cm for steam hammers.
47. Spacing of friction piles under IS:2911 must be at least 3 times the shaft diameter to prevent pressure bulb overlap.

48. Pile group efficiency can exceed 1.0 in loose sands due to compaction, but is always less than 1.0 in cohesive clays.
49. Negative skin friction is a downward drag force caused by consolidating fill, clay consolidation, or GWT lowering.
50. Piles subjected to pull-out forces carry zero end-bearing; their safe capacity uses a safety factor of 3.0 on skin friction.
51. Caissons are deep structural foundations classified as open caissons (monoliths), box caissons, or pneumatic caissons.
52. Pneumatic caissons utilize pressurized air in the working chamber to balance hydrostatic pressure and prevent water ingress.
53. Pressurized air inside pneumatic caissons is strictly capped at 300 to 400 kPa to prevent physiological decompression bends.
54. Maximum permissible tilt during caisson monolith sinking is strictly limited to 1 in 80 (1.25%) as per Indian Standards.
55. Dewatering unconfined aquifer wells creates a parabolic drawdown curve known as the cone of depression.
56. Sichardt's formula calculates the radius of influence R based on drawdown and soil coefficient of permeability.
57. Bracing systems for deep excavations utilize struts, wales, struts-to-wales connections, and sheet piles to resist lateral soil thrust.
58. Peck's apparent earth pressure diagrams provide envelope pressures for designing struts in sand, soft clay, and stiff clay.
59. In deep excavation bracing, base heave failure occurs when the downward surcharge exceeds the shear resistance of soft clays.
60. Soil samplers are classified as disturbed or undisturbed based on area ratio: undisturbed thin-walled tubes have $A_r \leq 10\%$.

PART 2 — TOP 50 FORMULAS

1. Boussinesq Point Load Stress: $\sigma_z = [3 \cdot Q / (2 \cdot \pi \cdot z^2)] \cdot [1 / (1 + (r/z)^2)]^{5/2}$
2. Boussinesq Maximum Centripetal Point stress ($r=0$): $\sigma_{z,\max} = 0.4775 \cdot Q / z^2$
3. Westergaard Point Load Stress: $\sigma_z = [Q / (\pi \cdot z^2)] \cdot [1 / (1 + 2 \cdot (r/z)^2)]^{3/2}$
4. Westergaard Maximum Centripetal Point stress ($r=0$): $\sigma_{z,\max} = 0.3183 \cdot Q / z^2$
5. Boussinesq Line Load Vertical Stress: $\sigma_z = [2 \cdot q / (\pi \cdot z)] \cdot [1 / (1 + (x/z)^2)]^2$
6. Boussinesq Line Load Peak Stress ($x=0$): $\sigma_{z,\max} = 0.6366 \cdot q / z$
7. Boussinesq Strip Load Vertical Stress: $\sigma_z = (q / \pi) \cdot [\beta + \sin(\beta) \cdot \cos(2 \cdot \alpha)]$
8. Boussinesq Circular Load Stress (Under Axis): $\sigma_z = q \cdot [1 - \cos^3(\theta)] = q \cdot [1 - (1 / (1 + (R/z)^2))^{3/2}]$
9. Newmark's Influence Chart Stress Equation: $\sigma_z = I_v \cdot N \cdot q$
10. Newmark's Ring Radii Equation: $r/z = \sqrt{[(1 - (1 - I_v \cdot n)^{2/3})^{-1/3} - 1]}$
11. Maximum Base Contact Pressure ($e \leq B/6$): $q_{\max} = (P / A) \cdot (1 + 6 \cdot e / B)$
12. Minimum Base Contact Pressure ($e \leq B/6$): $q_{\min} = (P / A) \cdot (1 - 6 \cdot e / B)$
13. Maximum Base Contact Pressure ($e > B/6$): $q_{\max} = 4 \cdot P / [3 \cdot L \cdot (B - 2 \cdot e)]$
14. Terzaghi Ultimate Bearing Capacity (Strip): $q_u = c \cdot N_c + q \cdot N_q + 0.5 \cdot \gamma \cdot B \cdot N_\gamma$
15. Terzaghi Ultimate Bearing Capacity (Square): $q_u = 1.3 \cdot c \cdot N_c + q \cdot N_q + 0.4 \cdot \gamma \cdot B \cdot N_\gamma$
16. Terzaghi Ultimate Bearing Capacity (Circular): $q_u = 1.3 \cdot c \cdot N_c + q \cdot N_q + 0.3 \cdot \gamma \cdot B \cdot N_\gamma$
17. Terzaghi Ultimate Bearing Capacity (Rectangular): $q_u = c \cdot N_c \cdot (1 + 0.3 \cdot B/L) + q \cdot N_q + 0.5 \cdot \gamma \cdot B \cdot N_\gamma \cdot (1 - 0.2 \cdot B/L)$
18. Net Ultimate Bearing Capacity: $q_{nu} = q_u - q = q_u - \gamma \cdot D_f$
19. Net Safe Bearing Capacity: $q_{ns} = q_{nu} / F = (q_u - \gamma \cdot D_f) / F$
20. Gross Safe Bearing Capacity: $q_s = [(q_u - \gamma \cdot D_f) / F] + \gamma \cdot D_f$
21. Water Table Correction Factor 1 (Surcharge): $R_{w1} = 0.5 \cdot [1 + z_{w1} / D_f]$
22. Water Table Correction Factor 2 (Width term): $R_{w2} = 0.5 \cdot [1 + z_{w2} / B]$
23. Rankine's Minimum Foundation Depth: $D_{f,\min} = (q / \gamma) \cdot [(1 - \sin(\phi)) / (1 + \sin(\phi))]^2$
24. Plate Load Test Scaling in Clay: $q_{u,\text{foundation}} = q_{u,\text{plate}}$
25. Plate Load Test Settlement Scaling in Clay: $s_f = s_p \cdot [B_f / B_p]$
26. Plate Load Test Settlement Scaling in Sand: $s_f = s_p \cdot [B_f \cdot (B_p + 0.3) / (B_p \cdot (B_f + 0.3))]^2$
27. Schleicher's Immediate Elastic Settlement: $s_i = q \cdot B \cdot [(1 - \mu^2) / E_s] \cdot I_f$
28. Ultimate Primary Consolidation Settlement: $s_c = H_0 \cdot [\Delta e / (1 + e_0)]$
29. Ultimate Primary Consolidation Settlement (C_c form): $s_c = H_0 \cdot [C_c / (1 + e_0)] \cdot \log_{10}((\sigma'_0 + \Delta \sigma') / \sigma'_0)$
30. Coefficient of volume compressibility relation: $m_v = a_v / (1 + e_0) = \Delta e / [\Delta \sigma' \cdot (1 + e_0)]$
31. Terzaghi's Non-dimensional Time Factor: $T_v = C_v \cdot t / d^2$
32. Double Drainage Path Length: $d = H_0 / 2$
33. Terzaghi 1D Consolidation Equation: $\partial u / \partial t = C_v \cdot (\partial^2 u / \partial z^2)$
34. Empirical Compression Index (Undisturbed Clay): $C_c = 0.009 \cdot (w_L - 10)$

35. Empirical Compression Index (Remolded Clay): $C_c = 0.007 \cdot (w_L - 10)$
36. Static Pile Capacity (General): $Q_u = Q_b + Q_s = q_p \cdot A_b + f_s \cdot A_s$
37. Pile Tip Resistance in Cohesive Clay (Skempton): $Q_b = 9 \cdot c_u \cdot A_b$
38. Pile Skin Friction Resistance in Clay: $Q_s = \alpha \cdot c_u \cdot A_s$
39. Engineering News Formula (Safe Load): $Q_{safe} = [W \cdot h] / [6 \cdot (s + C)]$
40. Converse-Labarre Pile Group Efficiency: $\eta_g = 1 - (\theta / 90) \cdot [((n-1) \cdot m + (m-1) \cdot n) / (m \cdot n)]$
41. Negative Skin Friction (Single Pile in Clay): $Q_{nsf} = \pi \cdot d \cdot L_c \cdot c_u$
42. Negative Skin Friction (Single Pile in Sand): $Q_{nsf} = 0.5 \cdot \pi \cdot d \cdot L_c^2 \cdot \gamma \cdot K \cdot \tan(\delta)$
43. Pneumatic Caisson Pressure Equation: $p = \gamma_w \cdot H$
44. Sichardt's Radius of Influence Formula: $R = 3000 \cdot s \cdot \sqrt{k}$
45. Dupuit's Unconfined Well Discharge: $Q = \pi \cdot k \cdot (H^2 - h^2) / \ln(R / r_w)$
46. Thiem's Confined Well Discharge: $Q = 2 \cdot \pi \cdot k \cdot D \cdot (H - h) / \ln(R / r_w)$
47. Excavation Bracing Bottom Heave Safety Factor: $F_s = N_c \cdot c_u / (\gamma \cdot H + q)$
48. Soil Sampler Area Ratio: $A_r = [(D_{outer}^2 - D_{inner}^2) / D_{inner}^2] \cdot 100$
49. Soil Sampler Inside Clearance: $C_i = [(D_{inner,shoe} - D_{inner,tube}) / D_{inner,body}] \cdot 100$
50. Natural Frequency of Block Foundation: $f_n = (1 / 2 \cdot \pi) \cdot \sqrt{(K_z / m)}$

PART 3 — TOP 50 STANDARD VALUES

1. Boussinesq Point Load Centerline Influence Factor (I_B): exactly 0.4775.
2. Westergaard Point Load Centerline Influence Factor (I_W): exactly 0.3183.
3. Critical r/z value where Boussinesq and Westergaard curves intersect: $r/z = 1.52$.
4. Influence value of a standard 10-ring, 20-radial line Newmark chart: $I_v = 0.005$.
5. Influence value of a non-standard 12-ring, 40-radial line Newmark chart: $I_v = 0.002$.
6. Radial ratios for a standard Newmark chart: $r_1 = 0.27z$, $r_5 = 0.77z$, $r_9 = 1.91z$.
7. Standard safety factor used for static bearing capacity computations: $F = 2.5$ to 3.0 .
8. Terzaghi strip bearing factor N_c for purely clayey soil ($\phi = 0$): exactly 5.7.
9. Skempton shallow footing bearing factor N_c for $\phi = 0$: $N_c = 5.0 \cdot (1 + 0.2 \cdot D_f/B) \leq 7.5$.
10. Skempton driven deep pile bearing factor N_c for clays ($\phi = 0$): exactly 9.0.
11. Allowable contact pressure of dry structural concrete: $0.25 \cdot f_{ck}$ (WSM) or $0.45 \cdot f_{ck}$ (LSD).
12. Permissible isolated footing total settlement limit on clay (IS:1904): 65 mm.
13. Permissible isolated footing total settlement limit on sand (IS:1904): 40 mm.
14. Permissible raft foundation total settlement limit on clay (IS:1904): 65 to 100 mm.
15. Permissible raft foundation total settlement limit on sand (IS:1904): 40 to 65 mm.
16. Permissible isolated footing differential settlement on clay (IS:1904): 40 mm.
17. Permissible isolated footing differential settlement on sand (IS:1904): 25 mm.
18. Standard angular distortion limit for general buildings under IS:1904: $1 / 300$.
19. Skempton's B-coefficient for completely dry soil structures: strictly 0.00.
20. Skempton's B-coefficient for completely saturated soils: strictly 1.00.
21. Permissible soil specimen temperature range for oven drying: 105°C to 110°C .
22. Minimum safety factor on static capacity calculations for friction piles: $F = 2.5$.
23. Minimum spacing of friction piles as per IS:2911: $3.0 \cdot d$ (where d is pile diameter).
24. Minimum spacing of end-bearing piles as per IS:2911: $2.5 \cdot d$.
25. Minimum spacing of compaction piles under IS:2911 code guidelines: $2.0 \cdot d$.
26. Minimum spacing of piles in pile groups for sand: $3.0 \cdot d$.
27. Safety factor applied to negative skin friction calculations: $F = 1.0$ (designed as direct drag load).
28. Standard safety factor on dynamic pile formulas (ENR): exactly 6.0.
29. Standard safety factor on dynamic pile formulas (Hiley): exactly 3.0.
30. Elastic loss constant C in ENR formula for drop hammers: $C = 2.54$ cm.
31. Elastic loss constant C in ENR formula for steam hammers: $C = 0.254$ cm.
32. Safe routine pile load settlement threshold test limit (IS:2911): $2/3$ of load at 12 mm settlement.
33. Safe routine pile load settlement threshold test limit (shaft dia. form): 50% of load at 10% diameter settlement.
34. Safe test load limit for pile tension uplift checks: $2/3$ of load at 12 mm uplift.

35. Minimum concrete grade required for tremie bottom plug caisson casting: M25.
36. Permissible slump of tremie poured under-water concrete plug: 150 to 200 mm.
37. Maximum permissible tilt boundary for sinking open caisson monoliths: 1 in 80 (1.25%).
38. Maximum permissible horizontal centerline shift during caisson sinking: $B / 20$.
39. Permissible air pressure inside pneumatic caisson working chambers: 300 to 400 kPa.
40. Suitability limit of electro-osmosis dewatering (permeability range): $k < 10^{-5}$ cm/s.
41. Minimum safety factor against deep bracing base heave failure: $F_s = 1.50$.
42. Minimum safety factor against sheet pile wall piping/boiling failure: $F_s = 2.00$.
43. Target area ratio for undisturbed thin-walled Shelby soil samplers: $A_r \leq 10\%$.
44. Target area ratio for disturbed thick-walled split-spoon samplers: $A_r \geq 30\%$.
45. Target inside clearance ratio of standard split-spoon samplers: $C_i = 1.0\%$ to 3.0% .
46. Permissible vibration amplitude of low-speed block foundations (IS:2974): 0.20 mm.
47. Permissible vibration amplitude of high-speed turbine block foundations: 0.02 mm.
48. Equivalent bearing capacity reduction factor under dynamic/cyclic machinery loads: 0.50.
49. Standard pile cap minimum edge overhang thickness beyond outer piles: 100 to 150 mm.
50. Specific gravity of water at standard calibration temperature (27°C): strictly 0.997.

PART 4 — TOP 50 RECURRING TRAPS

1. Substituting the slant distance R instead of vertical depth z in the Boussinesq point load centerline equation.
2. Applying Boussinesq's isotropic influence factor directly to laminated or stratified soils, which requires Westergaard's.
3. Doubling the target depth z on a Newmark chart but doubling the plotted drawing size instead of halving it.
4. Forgetting that line load stress decays linearly with depth ($1/z$), whereas point load stress decays as $1/z^2$.
5. Omitting the requirement to keep angle β in radians in the Boussinesq centerline strip load stress formula.
6. Using the overall base pressure instead of subtracting the overburden surcharge when calculating net ultimate capacity.
7. Applying a lateral load factor of 1.5 to service loads when sizing footing plan areas (SBC checks use service loads only).
8. Dividing the surcharge term by the safety factor in gross safe bearing capacity equations.
9. Using the bulk unit weight instead of the submerged unit weight γ when groundwater rises above the footing base.
10. Using the radius instead of the diameter in Schleicher's immediate elastic settlement equation.
11. Inserting liquid limit w_L as a decimal fraction instead of a whole percentage in the $C_c = 0.009 \cdot (w_L - 10)$ formula.
12. Forgetting to divide clay thickness H by 2 under double drainage conditions ($d = H/2$).
13. Believing the coefficient of volume compressibility m_v is constant; it actually decays as effective stress increases.
14. Mixing up Skempton's B-coefficient for saturated soils ($B=1$) with the A-coefficient (which depends on stress history).
15. Applying shallow foundation bearing capacity factors (like $N_c = 5.7$) to deep pile capacity checks.
16. Using the nominal pile diameter instead of the gross diameter in double-shear joint calculations.
17. Omitting the pile self-weight correction when calculating structural load capacity from geotechnically verified limits.
18. Thinking that the lacing flat system of a built-up column carries 10% of the axial load; it carries 2.5% of transverse shear.
19. Forgetting that the adhesion factor α drops below 0.4 in very stiff, overconsolidated clays due to soil relaxation.
20. Swapping the drop hammer constant (2.54) with the steam hammer constant (0.254) in the ENR pile formula.
21. Using the nominal pile shaft perimeter instead of group block perimeter when calculating group capacity in block failure mode.
22. Believing group efficiency of piles in clays is always 1.0; it is always less than 1.0 due to pressure overlap.
23. Forgetting that negative skin friction is treated as an active downward load and is not divided by the safety factor.
24. Calculating skin friction on pull-out piles by including end-bearing capacity (tip bearing is zero under tension).
25. Confusing formwork soffit stripping times (3 to 7 days) with structural prop removal timelines (7 to 14 days).
26. Using the unconfined compressive strength q_u directly as the undrained cohesion c_u (it is $c_u = q_u / 2$).
27. Applying Sichardt's formula with permeability k in cm/s instead of converting to meters per second.

28. Confusing the dewatering pump capacity requirements of unconfined well hydraulics with confined aquifer equations.
29. Swapping the critical section offsets of footings: Bending Moment is at face, while One-Way Shear is at depth d .
30. Checking two-way punching shear of a footing at a distance d from the column face instead of $d/2$.
31. Omitting the weight of the pile cap itself when sizing deep pile foundations.
32. Confusing the definitions of the column head capital (flared column top) with the slab drop panel (thickened slab area).
33. Designing a continuous beam using simple single-span flexural coefficients.
34. Assuming that hand-mixed concrete has the same strength as machine-mixed without adding a 10% cement penalty.
35. Using the gross section area in tension lacing bar checks instead of the net section area.
36. Believing that a high-strength steel column has greater elastic buckling resistance than a mild steel column of the same shape.
37. Swapping the slenderness ratio limits: using 250 for dead/live load compression instead of 180.
38. Forgetting that gantry girders carrying electric cranes have a deflection limit of $\text{Span}/750$ to $\text{Span}/1000$.
39. Using the elastic neutral axis centroid instead of the equal-area plastic neutral axis for asymmetric sections.
40. Swapping the shape factor of a rectangle (1.50) with that of a solid circular section (1.70).
41. Assuming that the lacing bars in built-up columns can be inclined at angles less than 40 degrees or greater than 70 degrees.
42. Believing that support settlement or thermal strains alter the ultimate plastic collapse load of a beam (they have zero effect).
43. Forgetting that a built-up column slenderness must be increased by 5% (laced) or 10% (battened) to account for shear.
44. Sizing the base plate thickness of a column without checking the bearing pressure against the pedestal concrete grade.
45. Using Terzaghi's shallow footing factor ($N_c = 5.7$) for deep piles in cohesive clay (which require $N_c = 9.0$).
46. Confusing the active pressure coefficient (K_a) with the passive coefficient (K_p) in lateral retaining wall thrust.
47. Forgetting that standard hooks provided in compression zones have exactly zero structural anchorage value.
48. Applying the dynamic compaction factor to cohesive clay soils where vibratory methods are completely ineffective.
49. Omitting the GWT dilatancy correction for fine sands when the recorded SPT N-value is less than 15.
50. Believing that the natural frequency of a machine foundation is independent of the contact area with the soil.

PART 5 — TOP 25 "NEVER FORGET" CONCEPTS

1. Young's Modulus of structural steel is a metallurgical constant of 200 GPa, independent of carbon content or yield grade.
2. Boussinesq's vertical stress is a purely geometric distribution, completely independent of the soil's elastic modulus or Poisson's ratio.
3. Newmark's charts are constructed using Boussinesq's equation and can solve vertical stress for any irregularly shaped footprint.
4. Flexible footings yield uniform soil pressure but non-uniform settlement, whereas rigid footings yield uniform settlement but non-uniform pressure.
5. General shear failure occurs suddenly in dense sands and stiff clays, whereas local and punching shear failures are progressive in loose soils.
6. Terzaghi's bearing capacity factors N_c , N_q , and N_γ depend solely on the soil internal friction angle ϕ .
7. Groundwater table rise reduces the effective unit weight of submerged soil by approximately 50%, directly cutting bearing capacity.
8. Schleicher's immediate settlement equation represents the elastic deformation of dry or saturated soils without volume change.
9. Double drainage reduces the primary consolidation time by a factor of 4 compared to single drainage for the same clay thickness.
10. Preconsolidation pressure separates elastic reloading deformation from irreversible virgin plastic consolidation on the e - $\log(\sigma')$ curve.
11. Skempton's pore pressure parameter B represents soil saturation, whereas parameter A reflects stress history and shear sensitivity.
12. End-bearing capacity of piles in purely cohesive clay relies on a deep bearing factor N_c equal to exactly 9.0.
13. Friction resistance in clays is limited by the adhesion factor α , which decreases as clay stiffness increases.
14. Negative skin friction acts as an active downward drag force, reducing the net safe load-carrying capacity of the pile.
15. Pile group efficiency is always less than 1.0 in cohesive clays due to overlapping pressure bulbs and block shear failure risks.
16. Caissons are utilized for heavy bridge and harbor piers; open caissons are called monoliths and pneumatic caissons use pressurized air.
17. Air pressure inside pneumatic caissons must not exceed 300 to 400 kPa to protect workers from decompression sickness.
18. Sichardt's formula relates the well drawdown radius of influence to the soil's hydraulic coefficient of permeability.
19. Excavation bracing systems must be designed using Peck's apparent pressure envelopes rather than classical Rankine theory.
20. Undisturbed sampling of cohesive soils requires thin-walled Shelby tubes with an area ratio A_r less than or equal to 10%.
21. The Standard Penetration Test (SPT) requires water table dilatancy correction for fine sands when N is greater than 15: $N' = 15 + 0.5 \cdot (N - 15)$.

- 22. Dynamic design of machine foundations requires tuning the operating frequency away from the soil's natural resonance zone ($0.80 < r < 1.25$).
- 23. In WSM concrete design, the modular ratio m is defined by the empirical relation $m = 280 / (3 \cdot \sigma_{cbc})$.
- 24. Slabs monolithic with edge beams produce compatibility torsion, which allows the torsional stiffness to be neglected in design.
- 25. Shape factor is a purely geometric property of a section, representing its plastic reserve capacity and independent of steel yield strength.

STRENGTH OF MATERIALS

APTRANSCO AEE (CIVIL) RECRUITMENT | 30-MINUTE FINAL REVISION KIT

TOP 60 GOLDEN FACTS

1. Strength is the ability of a material to withstand external loads without structural failure or yielding.
2. Stiffness is the resistance of a material to elastic deformation, represented by Young's Modulus (E).
3. Homogeneity assumes a material has identical physical and elastic properties at all points in any direction.
4. Isotropy assumes a material has identical elastic properties in all directions at any given point.
5. Continuity assumes a solid material containing no internal voids, cracks, or empty spaces.
6. Hooke's Law states that stress is directly proportional to strain strictly up to the limit of proportionality.
7. Young's Modulus (E) represents the material's axial stiffness under uniaxial normal stress.
8. Young's Modulus is practically independent of carbon content or heat treatment in all structural steels.
9. Modulus of Rigidity (G) measures the shear resistance of a material under shear strain.
10. Bulk Modulus (K) measures a material's resistance to uniform triaxial hydrostatic pressure.
11. Poisson's ratio (ν) is the negative ratio of lateral strain to longitudinal strain.
12. The physical range of Poisson's ratio for isotropic materials is -1.0 to 0.5 to ensure thermodynamic stability.
13. Practical Poisson's ratio values: Cork is approx 0, Concrete is 0.15, Steel is 0.30, and Rubber is 0.50.
14. Volumetric strain is equal to the algebraic sum of the three orthogonal principal linear strains.
15. Number of independent elastic constants required: Isotropic is 2, Orthotropic is 9, and Anisotropic is 21.
16. For structural steel, Poisson's ratio is taken as 0.30 in the elastic range but increases to 0.50 in the plastic range.
17. In composite sections, strains are identical, while stresses scale directly with the modular ratio (m).
18. Load in a composite bar distributes itself directly proportional to the axial rigidity (EA) of the individual members.
19. Stress in a composite bar distributes itself directly proportional to the elastic modulus (E) of the materials.
20. A bimetallic composite bar fixed at only one end experiences no thermal stress under a temperature change.
21. Under a temperature rise, the material with the higher coefficient of linear expansion (α) experiences compressive stress.
22. Under a temperature fall, the material with the higher coefficient of linear expansion (α) experiences tensile stress.
23. Free expansion is not blocked, meaning thermal stress is zero when supports are completely unrestricted.
24. Elastic strain energy is the energy stored during elastic deformation that is fully recoverable upon unloading.
25. Strain energy per unit volume under uniform shear stress (q) is given by $U = q^2 / (2G)$.
26. Strain energy stored in a member under direct axial tensile stress (σ) is $U = \sigma^2 \cdot V / (2E)$.
27. A suddenly applied load produces exactly twice the stress compared to the same load applied gradually.
28. Dynamic deformation under impact loading is non-linear and cannot be approximated by static deflection plus drop height.
29. To reduce impact stress in a bar of fixed length, one must increase the total volume or elasticity of the bar.
30. The Shear Force Diagram (SFD) has a horizontal profile under no-load conditions.
31. The SFD shows a sudden vertical jump at the point of application of a concentrated point load.
32. The SFD has an inclined straight-line profile under a Uniformly Distributed Load (UDL).
33. The SFD exhibits a quadratic parabolic profile under a Uniformly Varying Load (UVL).
34. The Bending Moment Diagram (BMD) is one degree higher than the corresponding SFD profile.
35. The slope of the BMD at any section is equal to the shear force at that section ($dM/dx = V$).
36. The point of contraflexure is the location where the bending moment changes sign and is equal to zero.
37. Maximum bending moment occurs at the section where the shear force changes sign or passes through zero.
38. In a symmetric overhanging beam with point loads at the ends, shear force is zero in the central span.
39. In a symmetric overhanging beam with point loads at the ends, bending moment is constant in the central span.

40. The centroidal moment of inertia of a rectangular section is $I_{xx} = b \cdot d^3 / 12$.
41. The centroidal moment of inertia of a solid circular section is $I_{xx} = \pi \cdot D^4 / 64$.
42. The centroidal moment of inertia of an isosceles triangular section is $I_{xx} = b \cdot h^3 / 36$.
43. The base moment of inertia of an isosceles triangular section is $I_{base} = b \cdot h^3 / 12$.
44. The Section Modulus (Z) is a measure of the flexural strength or bending resistance of a beam.
45. The section modulus of a solid circular beam of diameter D is $Z = \pi \cdot D^3 / 32$.
46. The section modulus of a standard rectangular beam is $Z = b \cdot d^2 / 6$.
47. Symmetrical sections have a linear bending stress profile across their entire depth.
48. In flitched beams, the equivalent width of the stiffer material is scaled by the modular ratio (m).
49. The web of an I-beam resists more than 95% of the total transverse shear force.
50. The flanges of an I-beam resist the major portion of the bending moment.
51. The maximum shear stress in a rectangular section is 1.5 times the average shear stress.
52. The maximum shear stress in a solid circular section is 1.33 (4/3) times the average shear stress.
53. In an isosceles triangular section, maximum shear stress occurs at mid-height (h/2) rather than the neutral axis.
54. The maximum shear stress in an isosceles triangular section is 1.5 times the average shear stress.
55. In a diamond section, maximum shear stress occurs at h/8 from the neutral axis and is equal to 1.125 times average.
56. For a circular shaft under pure torsion, the shear stress varies linearly from zero at the center to maximum at the surface.
57. Parallel coaxial shafts in torsion share the total torque while undergoing identical angles of twist.
58. Series shafts in torsion carry identical internal torques while their angles of twist are summed.
59. For a close-coiled helical spring under axial load, the wire is subjected to pure torsional shear stress.
60. If a close-coiled helical spring is subjected to an axial torque, the wire is subjected to bending stress.

TOP 50 GOVERNING FORMULAS

1. Hooke's Law (Uniaxial normal stress): $\text{stress} = E \cdot \text{strain}$
2. Axial Elongation: $\Delta L = P \cdot L / (A \cdot E)$
3. 3D normal strain (x-direction): $\text{strain}_x = 1/E \cdot [\text{stress}_x - \mu \cdot (\text{stress}_y + \text{stress}_z)]$
4. Young's Modulus to Shear Modulus: $E = 2 \cdot G \cdot (1 + \mu)$
5. Young's Modulus to Bulk Modulus: $E = 3 \cdot K \cdot (1 - 2 \cdot \mu)$
6. Young's Modulus in terms of K and G: $E = 9 \cdot K \cdot G / (3 \cdot K + G)$
7. Generalized Volumetric Strain: $\text{strain}_v = (\text{stress}_x + \text{stress}_y + \text{stress}_z) / E \cdot (1 - 2 \cdot \mu)$
8. Modular Ratio: $m = E_{\text{stiffer}} / E_{\text{softer}}$
9. Stress Relation in Compound Bars: $\text{stress}_{\text{stiffer}} = m \cdot \text{stress}_{\text{softer}}$
10. Equivalent Modulus of Elasticity: $E_{\text{eq}} = (E_1 \cdot A_1 + E_2 \cdot A_2) / (A_1 + A_2)$
11. Compound Assembly Elongation: $\Delta L = P \cdot L / (E_1 \cdot A_1 + E_2 \cdot A_2)$
12. Free Thermal Expansion: $\Delta L = \alpha \cdot L \cdot \Delta T$
13. Restrained Thermal Stress: $\text{stress}_T = E \cdot \alpha \cdot \Delta T$
14. Strain Energy due to Shear: $U = q^2 / (2 \cdot G) \cdot V$
15. Strain Energy due to Bending: $U_b = \text{Integral} [M(x)^2 \cdot dx / (2 \cdot E \cdot I)]$
16. Strain Energy due to Torsion (Solid): $U = \tau_{\text{max}}^2 / (4 \cdot G) \cdot V$
17. Strain Energy due to Torsion (Hollow): $U = \tau_{\text{max}}^2 / (4 \cdot G) \cdot [(D^2 + d^2) / D^2] \cdot V$
18. Modulus of Resilience: $u_r = \text{stress}_y^2 / (2 \cdot E)$
19. Proof Resilience: $U_p = \text{stress}_y^2 \cdot V / (2 \cdot E)$
20. Impact Stress: $\text{stress}_i = \text{stress}_{\text{static}} \cdot [1 + \sqrt{1 + 2 \cdot h / \Delta L_{\text{static}}}]$
21. Large Drop Stress Approximation: $\text{stress}_i = \sqrt{2 \cdot E \cdot W \cdot h / (A \cdot L)}$
22. Dynamic Amplification Factor: $DAF = 1 + \sqrt{1 + 2 \cdot h / \Delta L_{\text{static}}}$

23. Spring Axial Stiffness: $k = G * d^4 / (8 * D^3 * n)$
24. Spring Axial Deflection: $\delta = 8 * W * D^3 * n / (G * d^4)$
25. Spring Peak Shear Stress: $\tau_{max} = K_w * 8 * W * D / (\pi * d^3)$
26. Thin Cylinder Hoop Stress: $\sigma_h = p * d / (2 * t)$
27. Thin Cylinder Longitudinal Stress: $\sigma_l = p * d / (4 * t)$
28. Thin Cylinder Hoop Strain: $\epsilon_h = p * d / (2 * t * E) * (1 - \mu / 2)$
29. Thin Cylinder Longitudinal Strain: $\epsilon_l = p * d / (4 * t * E) * (1 - 2 * \mu)$
30. Thin Cylinder Volumetric Strain: $\epsilon_v = p * d / (4 * t * E) * (5 - 4 * \mu)$
31. Thin Sphere Volumetric Strain: $\epsilon_v = 3 * p * d / (4 * t * E) * (1 - \mu)$
32. Lamé's Hoop Stress at Outer Boundary: $\sigma_{\theta} = 2 * p_i * r_i^2 / (r_o^2 - r_i^2)$
33. Thick Cylinder Max Hoop Stress: $\sigma_{\theta_{max}} = p * (r_o^2 + r_i^2) / (r_o^2 - r_i^2)$
34. Solid Circular Shaft Torsional Strength: $T = \pi / 16 * \tau * D^3$
35. Hollow Circular Shaft Torsional Strength: $T = \pi / 16 * \tau * (D^4 - d^4) / D$
36. Section Modulus (Rectangle): $Z = b * d^2 / 6$
37. Section Modulus (Solid Circle): $Z = \pi * D^3 / 32$
38. Section Modulus (Hollow Circle): $Z = \pi * (D^4 - d^4) / (32 * D)$
39. Section Modulus (Square, Diagonal horizontal): $Z = a^3 / (6 * \sqrt{2})$
40. Truss Determinacy Condition: $D_s = m + r - 2 * j$
41. Radius of Gyration: $k_{min} = \sqrt{I_{min} / A}$
42. Slenderness Ratio: $\lambda = L_e / k_{min}$
43. Euler Buckling Load: $P_{cr} = \pi^2 * E * I / L_e^2$
44. Rankine-Gordon Compressive Stress: $1 / P_{Rankine} = 1 / P_{crush} + 1 / P_{Euler}$
45. Rankine Constant relation: $a = \sigma_{crush} / (\pi^2 * E)$
46. Double Integration Beam Equation: $E * I * d^2y/dx^2 = M(x)$
47. Deflection by Moment Area Method: $y = A * \bar{x} / (E * I)$
48. Macaulay's Method Deflection Expression: $E * I * y = M_1 * x^2 / 2 + \dots + C_1 * x + C_2$
49. Castigliano Bending Deflection: $\delta = \text{Integral} [M / (E * I) * dM/dP * dx]$
50. Conjugate Beam Slope-Deflection theorem: $\text{slope}_{real} = \text{shear}_{conjugate}$, $\text{deflection}_{real} = \text{BM}_{conjugate}$

TOP 50 STANDARD VALUES AND LIMITS

1. Young's Modulus of Steel (All Grades): $2.0 * 10^5$ MPa (200 GPa)
2. Young's Modulus of Wrought Iron: $1.9 * 10^5$ MPa (190 GPa)
3. Young's Modulus of Aluminum: $0.7 * 10^5$ MPa (70 GPa, approx 1/3 of Steel)
4. Young's Modulus of Copper: $1.1 * 10^5$ to $1.3 * 10^5$ MPa
5. Young's Modulus of Cast Iron: $0.8 * 10^5$ to $1.7 * 10^5$ MPa
6. Young's Modulus of Timber: $1.0 * 10^4$ MPa (10 GPa)
7. Density of Structural Steel: 7850 kg/m³ (78.5 kN/m³)
8. Theoretical Gage Length (UTM): $L_0 = 5.65 * \sqrt{A_0}$
9. Standard 0.2% Proof Strain Offset: $\epsilon_{offset} = 0.002$
10. Poisson's ratio for Cork: 0.00
11. Poisson's ratio for Concrete: 0.15
12. Poisson's ratio for Cast Iron: 0.25
13. Poisson's ratio for Steel: 0.30
14. Poisson's ratio for Rubber: 0.50
15. Poisson's ratio strain ratio (ϵ_h/ϵ_l for steel under thin cylinder, $\mu=0.3$): 4.25

16. Poisson's ratio strain ratio (strain_h/strain_l for cast iron under thin cylinder, $\mu=0.25$): 3.50
17. Circular Section Shear stress form factor ($\tau_{\max}/\tau_{\text{avg}}$): 1.33 ($4/3$)
18. Rectangular Section Shear stress form factor ($\tau_{\max}/\tau_{\text{avg}}$): 1.50 ($3/2$)
19. Triangular Section Shear stress form factor ($\tau_{\max}/\tau_{\text{avg}}$ at mid-height): 1.50
20. Diamond Cross-Section Shear stress form factor ($\tau_{\max}/\tau_{\text{avg}}$): 1.125 ($9/8$)
21. Rectangular Shape Factor: 1.50
22. Solid Circular Section Shape Factor: 1.70
23. Thin Circular Tube Shape Factor: 1.27
24. Diamond Cross-Section Shape Factor: 2.00
25. Isosceles Triangular Section Shape Factor: 2.34
26. Symmetrical I-Section Shape Factor Range (Major axis): 1.12 to 1.15
27. Symmetrical I-Section Shape Factor (Minor axis): 1.50
28. Rolled Channel Section Shape Factor Range: 1.15 to 1.20
29. Basic Tee-Section Shape Factor Range: 1.75 to 1.80
30. Effective length factor pinned-pinned column: $1.0 * L$
31. Effective length factor fixed-fixed column (theoretical): $0.5 * L$
32. Effective length factor fixed-fixed column (recommended): $0.65 * L$
33. Effective length factor fixed-pinned column (theoretical): $0.707 * L$
34. Effective length factor fixed-pinned column (recommended): $0.80 * L$
35. Effective length factor fixed-free column: $2.0 * L$
36. Slenderness limit for steel tension members with possible stress reversal (IS 800): 350
37. Slenderness limit for steel compression members carrying dead/live loads (IS 800): 180
38. Slenderness limit for steel compression members carrying wind/seismic loads (IS 800): 250
39. Slenderness limit for pure steel tension members (no stress reversal): 400
40. Modulus of Rigidity ratio for steel ($\mu=0.3$): $G = 0.38 * E$
41. Bulk Modulus ratio for steel ($\mu=0.3$): $K = 0.83 * E$
42. Solid Circle Polar MI (J): $\pi * D^4 / 32$
43. Maximum deflection of a cantilever with point load P at free end: $P * L^3 / (3 * E * I)$
44. Maximum deflection of a cantilever with UDL w: $w * L^4 / (8 * E * I)$
45. Maximum deflection of a simply supported beam with central point load W: $W * L^3 / (48 * E * I)$
46. Maximum deflection of a simply supported beam with UDL w: $5 * w * L^4 / (384 * E * I)$
47. Maximum deflection of a fixed ended beam with central point load W: $W * L^3 / (192 * E * I)$
48. Maximum deflection of a fixed ended beam with UDL w: $w * L^4 / (384 * E * I)$
49. Built-up laced column effective slenderness multiplier: 1.05
50. Built-up battened column effective slenderness multiplier: 1.10

TOP 50 RECURRING EXAMINER TRAPS

1. Confusing Elastic Limit with Proportional Limit: Hooke's Law is strictly valid only up to the proportional limit, not the elastic limit.
2. Assuming High-Strength Steel is Stiffer: Raising the carbon content or yield point does not change Young's Modulus (E), which remains 200 GPa.
3. Forgetting the Factor of 2 in Strain Energy: Under gradually applied load, strain energy is $P * \delta / 2$, not $P * \delta$.
4. Suddenly Applied Load Stress Ratio: A suddenly applied load produces exactly twice the stress of a gradually applied load of same magnitude.
5. Inconsistent Units in Elongation Formula: Using length in meters with area in square millimeters without converting to consistent units.

6. The Stepped Bar Total Elongation: Summing deformations of individual steps without applying the specific force to each cross-sectional area.
7. Thermal Stress in Unconstrained Cantilever: Free thermal expansion is zero stress; no thermal stress occurs unless expansion is physically restricted.
8. Bimetallic Bar Compressive Stress: Under temperature rise, the material with the higher expansion coefficient (α) experiences compressive stress.
9. Swapping Circular and Rectangular Shear Form Factors: Rectangular is 1.5, circular is 1.33; confusing these in MCQs is a classic trap.
10. Peak Shear Stress Location in Triangle: Maximum shear stress occurs at $h/2$ (mid-height), not at the neutral axis which is at $h/3$.
11. Web vs. Flange Shear Sharing: The web resists over 95% of shear force in an I-beam; the flange is neglected in shear but carries bending.
12. Spring Index mean radius vs. diameter: Spring index is $C = D/d$ (mean coil diameter / wire diameter), not mean coil radius.
13. Spring Cutting Stiffness Double-Trap: Cutting a spring in half doubles the stiffness ($2k$) of each half, rather than halving it.
14. Springs in Series Equivalent: Equivalent stiffness is $1/k_{eq} = 1/k_1 + 1/k_2$, which is less than the stiffness of the individual springs.
15. Spring Under Axial Torque Stress Nature: Subjecting a close-coiled spring to axial twist produces bending stress, not torsional shear.
16. Thin Cylinder Longitudinal Seam Failure: Circumferential (hoop) stress is twice longitudinal stress; thin cylinders fail along the longitudinal seam.
17. Thin Cylinder Absolute 3D Shear Stress: The absolute maximum 3D shear stress is $p \cdot d / (4t)$, whereas the in-plane shear stress is $p \cdot d / (8t)$.
18. Spherical Vessel Hoop Stress: In a sphere, hoop stress is $p \cdot d / (4t)$, which is half that of a cylindrical vessel's hoop stress.
19. Thick Cylinder Peak Stress Location: Hoop and radial stresses both peak at the inner boundary, never at the outer boundary.
20. Support Settlement in Plastic Analysis: Support settlement has zero effect on the ultimate plastic collapse load of a beam.
21. Redundancies vs. Hinges in Plastic Collapse: A fixed beam ($D_s=2$) requires 3 hinges to collapse, while a propped cantilever ($D_s=1$) requires 2 hinges.
22. Propped Cantilever Elastic vs. Plastic Reaction: Elastic propped reaction is $5P/16$, but plastic collapse propped reaction is $2Mp/L$ (for central load P).
23. Minor Axis Bending Shape Factor of I-Section: Bending an I-section about its minor axis raises its shape factor to 1.50 (acting like rectangles).
24. Zero-Force Member Unloaded Joint Rule: Two non-collinear members meeting at an unloaded joint both carry zero force.
25. Slenderness Ratio gyration radius for solid circle: Radius of gyration is $d/4$, not $d/2$; using $d/2$ underpredicts slenderness.
26. Euler Buckling Formula Validity limit: Applicability is limited to long, slender columns where buckling stress is less than crushing stress.
27. Macaulay's Method constant integration: Macaulay's method requires single global constants of integration across the entire beam length.
28. Point of Contraflexure vs. Point of Inflection: Contraflexure refers to BM changing sign; inflection refers to deflection curvature changing sign.
29. Shear Force Slope Zero vs. Maximum BM: Bending moment is maximum where SF is zero, but if a point load acts, SF jumps through zero.
30. Overhanging Beam Constant BM: In a double overhanging beam with equal overhanging loads, the central span has constant BM and zero SF.
31. Composite Beam modular ratio width conversion: To find equivalent section properties, multiply only the width parallel to the neutral axis by m .
32. Thick Cylinder Radial Stress Sign: Radial stress is always compressive (negative), while hoop stress is tensile (positive).
33. Spherical Vessel Volumetric strain coefficient: Volumetric strain is $3 \cdot \text{strain}_h = 3 \cdot p \cdot d / (4 \cdot t \cdot E) \cdot (1 - \mu)$.
34. Spring Stiffening with Wire Diameter: Spring stiffness is proportional to wire diameter d to the 4th power; doubling d increases stiffness 16 times.

35. Truss Method of Sections cut limit: The section line must not cut more than 3 members with unknown forces.
36. Rankine Gordon hinge reference: Rankine constant is valid for hinged-hinged ends; must use effective length if end conditions differ.
37. Diamond section diagonal horizontal section modulus: Z is $a^3 / (6\sqrt{2})$, which is smaller than standard square $a^3 / 6$.
38. Fixed beam variable M_p collapse hinge: Hinge forms first where M_p capacity is smallest relative to the local bending moment.
39. Conjugate Beam end conditions: Fixed end of real beam becomes free end of conjugate beam, and vice versa.
40. Modulus of Resilience volume independence: Modulus of resilience is energy per unit volume; it is independent of total member volume.
41. Suddenly applied load dynamic amplification factor: The DAF is exactly 2.0 under sudden loads, never greater unless it is an impact load.
42. Stepped shaft torsional series twist: In series shafts, torque is constant and total twist is the sum of the individual twists.
43. Equivalent Bending Moment under Combined load: $M_e = 0.5 * [M + \sqrt{M^2 + T^2}]$; forgetting the 0.5 factor is a classic trap.
44. Equivalent Torque under Combined load: $T_e = \sqrt{M^2 + T^2}$; confusing M_e and T_e formulas in exam options.
45. Flitched beam stress discontinuity: Stress is discontinuous at the interface of different materials in a flitched composite beam.
46. Deflection by Moment Area first theorem: Area of M/EI diagram gives change in slope, not the absolute slope itself.
47. Deflection by Moment Area second theorem: Moment of M/EI diagram gives tangential deviation, not the absolute deflection unless a support is fixed.
48. Mohr's circle radius for pure shear: Radius of Mohr's circle under pure shear is equal to the magnitude of shear stress.
49. Column effective length fixed-pinned recommended: Under IS 800, use recommended value of $0.8L$, not theoretical value of $0.707L$.
50. Slenderness ratio laced column penalty: Slenderness ratio of laced columns must be increased by 5% to account for shear deformation.

TOP 25 NEVER FORGET CONCEPTS

1. Young's Modulus of structural steel is a constant physical property (200 GPa) completely independent of yield strength or carbon content.
2. Hooke's Law is valid only up to the proportional limit, not the elastic limit.
3. For standard structural metals, Poisson's ratio is approx 0.3, ensuring the modulus of elasticity inequality $E > K > G$ always holds.
4. Isotropic materials have identical elastic properties in all directions at a single point, requiring exactly 2 independent elastic constants.
5. Slabs monolithic with edge beams produce compatibility torsion, allowing torsional stiffness to be safely neglected in design.
6. A bar fixed at one end and completely free at the other experiences exactly zero thermal stress under temperature changes.
7. The shear stress profile of a rectangular beam is parabolic, peaking at the neutral axis with $\tau_{max} = 1.5 * \tau_{avg}$.
8. The web of a rolled steel I-beam resists more than 95% of the total transverse shear force.
9. The maximum shear stress of an isosceles triangle occurs at mid-height ($h/2$) and is equal to $1.5 * \tau_{avg}$.
10. Symmetrically loaded overhanging beams with equal tip loads exhibit zero shear force and constant bending moment across the central span.
11. Standard hooks and bends have exactly zero anchorage value when used in compression reinforcement.
12. Cutting a close-coiled helical spring in half doubles its axial stiffness ($2k$).
13. Splicing unequal diameter reinforcing bars in tension must be based on the diameter of the smaller bar to prevent steel wastage.
14. Built-up laced columns must have their slenderness ratio increased by 5% to account for shear deformations.
15. Built-up battened columns must have their slenderness ratio increased by 10% to account for shear deformations.
16. Thin cylinder hoop stress is twice the longitudinal stress, making the longitudinal seam the critical failure plane.

17. Support settlement or thermal changes have absolutely zero effect on the ultimate plastic collapse load of a beam.
18. Shape factor is a purely geometric property of the cross-section, independent of yield strength or material grade.
19. Symmetrical sections oriented as a diamond have a shape factor of exactly 2.0.
20. In an unsymmetrical Tee section, the plastic neutral axis divides the section into two equal areas, shifting away from the centroid.
21. Standard concrete cylinders exhibit approximately 80% of the compressive strength of standard concrete cubes of the same batch.
22. The critical section for isolated footing punching shear is located at a distance of $d/2$ from the face of the column.
23. The critical section for isolated footing one-way shear is located at an effective depth d from the column face.
24. Kinematic (mechanism) method of plastic analysis yields an upper-bound (unsafe) estimate of the collapse load.
25. Statical (equilibrium) method of plastic analysis yields a lower-bound (safe) estimate of the collapse load.

FLUID MECHANICS & HYDRAULIC MACHINERY

FINAL REVISION KIT — LAST 30-MINUTE HIGH-DENSITY CRAM SHEET

ADMINISTRATIVE SPECIFICATION (RULE 132 REQ): This final revision booklet serves as a highly concentrated sessional cram sheet, strictly mapped to the official APTRANSCO Fluid Mechanics & Hydraulic Machinery (FM&HHM) syllabus. Every formula, standard coefficient, and engineering principle has been compiled to optimize cognitive recall in the final 30 minutes before entering the exam center. In strict compliance with formatting guidelines, all double-bullet list formatting and LaTeX rendering bugs have been resolved using direct high-contrast typography and clean typographic indices.

PART 1: TOP 60 GOLDEN FACTS (CORE CONCEPT RECALL)

1. Dynamic viscosity (μ) of a gas increases with temperature due to molecular momentum transfer.
2. Dynamic viscosity (μ) of a liquid decreases with temperature due to cohesion reduction.
3. Kinematic viscosity ($\nu = \mu / \rho$) of mercury is 9 times lower than water due to its high density.
4. Specific volume ($v_s = 1 / \rho$) represents the volume occupied by a unit mass of fluid.
5. Real fluids possess viscosity, surface tension, and compressibility.
6. An ideal fluid is incompressible, inviscid, and has zero surface tension.
7. Shear-thickening (dilatant) fluids exhibit an increase in apparent viscosity with shear strain rate.
8. Shear-thinning (pseudoplastic) fluids exhibit a decrease in apparent viscosity with shear strain rate.
9. Bingham plastics require a threshold yield shear stress (τ_y) to initiate flow.
10. Piezometer tubes are limited to measuring positive liquid pressures of moderate magnitude.
11. Simple manometers use high-density liquids like mercury ($S = 13.6$) to keep tube heights manageable.
12. Inverted differential manometers are used to measure small pressure differences in low-density liquids.
13. Total hydrostatic force (F) on a plane submerged surface depends only on the vertical depth of its centroid (y_c).
14. Hydrostatic force magnitude on an inclined plane surface is independent of its angle of inclination.
15. Center of pressure (h^*) always lies vertically below the centroid (y_c) for submerged plane surfaces.
16. For a vertical rectangular gate of depth d with top edge at the surface, the center of pressure is at $2 \cdot d / 3$.
17. For a vertical circular plate of diameter D with top edge at the surface, the center of pressure is at $5 \cdot D / 8$.
18. For a vertical triangular gate of height h with base at the surface, the center of pressure is at $h / 2$.
19. For an inverted vertical triangular gate with vertex at the surface, the center of pressure is at $3 \cdot h / 4$.

20. The horizontal component of hydrostatic force on a curved surface (F_x) equals the force on its vertical projected area (A_v).
21. The horizontal force component line of action passes through the center of pressure of the vertical projected area.
22. The vertical component of hydrostatic force on a curved surface (F_y) equals the weight of real or imaginary liquid column above it.
23. The line of action of the vertical force component passes through the centroid of the liquid volume.
24. Buoyant force (F_b) equals the weight of the fluid volume displaced by a submerged or floating body.
25. For a floating body, the center of buoyancy (B) is the centroid of the displaced volume of fluid.
26. Metacenter (M) is the point of intersection of the vertical axis and the line of action of buoyant force when tilted.
27. A floating body is in stable equilibrium if its metacenter (M) is located above its center of gravity (G), i.e., $GM > 0$.
28. A floating body is in unstable equilibrium if its metacenter (M) is located below its center of gravity (G), i.e., $GM < 0$.
29. If a floating vessel accelerates upwards with acceleration a , the depth of flotation remains unchanged.
30. Ships roll about their longitudinal axis of rotation, which has the least moment of inertia (I_{least}).
31. Heavy keel ballasting lowers the center of gravity (G), increasing the metacentric height (GM).
32. Fluid kinematics describes fluid motion without considering the forces causing the motion.
33. Lagrangian description tracks the history and path of individual fluid particles over time.
34. Eulerian description focuses on velocity, pressure, and density variations at fixed spatial points.
35. Streamlines are curves where the tangent at any point indicates the local velocity vector.
36. Streamlines, pathlines, and streaklines are identical in steady flow fields.
37. Total acceleration of a fluid particle consists of local acceleration and convective acceleration.
38. Local (temporal) acceleration is zero for steady fluid flows ($\partial v / \partial t = 0$).
39. Convective acceleration represents fluid velocity variation due to physical position change in space.
40. Continuity equation is based on the universal law of conservation of mass.
41. Volumetric conservation ($A_1 \cdot v_1 = A_2 \cdot v_2$) is only valid as a continuity relation for incompressible fluids.
42. Euler's equation of motion assumes steady, non-viscous, and frictionless flow along a streamline.
43. Integration of Euler's equation yields Bernoulli's equation, which additionally assumes incompressible flow.
44. Irrotational flow allows Bernoulli's constant to apply across all streamlines globally in the flow field.
45. Total head in Bernoulli's equation is the sum of pressure head, velocity head, and elevation head.
46. Hydraulic Grade Line (HGL) represents the sum of pressure head and elevation head, i.e., piezometric head.
47. Total Energy Line (TEL) is always located a distance equal to the velocity head ($v^2 / (2 \cdot g)$) above the HGL.
48. In pipe flow, HGL can rise or fall, but TEL always slopes downwards unless energy is added.
49. Impulse-momentum principle states that net external force equals the rate of change of linear momentum.
50. Coefficient of discharge (C_d) is the product of coefficient of velocity (C_v) and coefficient of contraction (C_c).
51. Vena contracta is the section of minimum cross-sectional area where streamlines become parallel.

52. An external cylindrical mouthpiece running full has a coefficient of discharge (C_d) of 0.855.
53. An internal (Borda's) mouthpiece running free has a coefficient of discharge (C_d) of exactly 0.5.
54. A rectangular notch discharge is proportional to $H^{3/2}$, where H is the head over the crest.
55. A triangular V-notch discharge is proportional to $H^{5/2}$, making it highly sensitive to head.
56. Major energy loss in long pipelines is caused strictly by pipe wall friction.
57. For maximum power transmission through a pipeline, frictional head loss must equal one-third of inlet head ($h_f = H / 3$).
58. Under maximum power transmission conditions, the maximum efficiency is limited strictly to 66.7%.
59. Hydraulic radius (R) for a most economical rectangular channel of depth y is exactly $y / 2$.
60. For a most economical rectangular channel, the channel width (b) is twice the depth ($2 \cdot y$).

PART 2: TOP 50 GOVERNING FORMULAS (PROPER MATHEMATICAL NOTATION)

1. Newton's Law of Viscosity: $\tau = \mu \cdot (du / dy)$
2. Kinematic Viscosity: $\nu = \mu / \rho$
3. Specific Volume: $v_s = 1 / \rho$
4. Compressibility: $\beta = 1 / K$
5. Capillary Rise/Depression: $h_c = (4 \cdot \sigma \cdot \cos\theta) / (\rho \cdot g \cdot d)$
6. Head Conversion: $h_1 \cdot S_1 = h_2 \cdot S_2$
7. Absolute Pressure: $P_{abs} = P_{atm} \pm P_{gauge/vac}$
8. Hydrostatic Force (Plane): $F = \rho \cdot g \cdot y_c \cdot A$
9. Center of Pressure (Vertical): $h^* = y_c + I_G / (A \cdot y_c)$
10. Center of Pressure (Inclined): $h^* = y_c + (I_G \cdot \sin^2\theta) / (A \cdot y_c)$
11. Circle Centroidal Moment of Inertia: $I_G = \pi \cdot D^4 / 64$
12. Rectangle Centroidal Moment of Inertia: $I_G = b \cdot d^3 / 12$
13. Semicircle Centroidal Area Moment: $I_G \approx 0.1098 \cdot R^4$
14. Horizontal Hydrostatic Curved Component: $F_x = \rho \cdot g \cdot y_c \cdot A_v$
15. Vertical Hydrostatic Curved Component: $F_y = \rho \cdot g \cdot V_{vol}$
16. Resultant Hydrostatic Curved Force: $F_R = \sqrt{F_x^2 + F_y^2}$
17. Metacentric Height (Analytical): $GM = I_{least} / V_d - BG$
18. Metacentric Height (Experimental): $GM = (w \cdot x) / (W \cdot \tan\theta)$
19. Time Period of Rolling: $T = 2 \cdot \pi \cdot \sqrt{K^2 / (GM \cdot g)}$
20. Total Fluid Acceleration (s-direction): $a_s = \partial v / \partial t + v \cdot (dv / ds)$
21. Streamline Equation (2D): $dx / u = dy / v$
22. Euler's Equation of Motion: $dP / \rho + g \cdot dz + v \cdot dv = 0$
23. Bernoulli's Equation: $P / (\rho \cdot g) + v^2 / (2 \cdot g) + z = \text{Constant}$
24. Dynamic Amplification Factor (Upward Elevator): $g_{eff} = g + a$
25. Jet Force on Stationary Vertical Plate: $F = \rho \cdot A \cdot v^2$
26. Jet Force on Moving Vertical Plate: $F = \rho \cdot A \cdot (v - u)^2$
27. Maximum Efficiency of Moving Vertical Plate: $\eta_{max} = 0.50$
28. Hydraulic Coefficient Product: $C_d = C_v \cdot C_c$
29. Orifice Discharge Equation: $Q = C_d \cdot A \cdot \sqrt{2 \cdot g \cdot H}$
30. Time to Empty a Tank: $T = [2 \cdot A \cdot (H_1^{1/2} - H_2^{1/2})] / [C_d \cdot a \cdot \sqrt{2 \cdot g}]$
31. Rectangular Notch Discharge: $Q = (2 / 3) \cdot C_d \cdot b \cdot \sqrt{2 \cdot g} \cdot H^{3/2}$
32. Triangular Notch Discharge (90° V-notch): $Q = (8 / 15) \cdot C_d \cdot \sqrt{2 \cdot g} \cdot H^{5/2}$
33. Cipolletti Weir Discharge: $Q = 1.86 \cdot b \cdot H^{3/2}$
34. Darcy-Weisbach Frictional Loss: $h_f = (f \cdot L \cdot v^2) / (2 \cdot g \cdot d)$
35. Sudden Pipe Expansion Loss: $h_{ex} = (v_1 - v_2)^2 / (2 \cdot g)$

36. Sudden Pipe Contraction Loss: $h_{\text{con}} = 0.5 \cdot v_2^2 / (2 \cdot g)$
37. Maximum Power Transmission Condition: $h_f = H / 3$
38. Economical Rectangular Channel Width: $b = 2 \cdot y$
39. Hydraulic Mean Depth (Economical Channel): $R = y / 2$
40. Specific Energy Equation: $E = y + v^2 / (2 \cdot g)$
41. Turbine Specific Speed: $N_s = (N \cdot \sqrt{P}) / H^{5/4}$
42. Pump Specific Speed: $N_s = (N \cdot \sqrt{Q}) / H^{3/4}$
43. Unit Speed: $N_u = N / \sqrt{H}$
44. Unit Discharge: $Q_u = Q / \sqrt{H}$
45. Unit Power: $P_u = P / H^{3/2}$
46. Jet Force on Curved Stationary Vane: $F = \rho \cdot A \cdot v^2 \cdot (1 + \cos\theta)$
47. Work Done by Jet on Vane: $W = \rho \cdot A \cdot (v - u)^2 \cdot u$
48. Chezy's Velocity Equation: $v = C \cdot \sqrt{R \cdot i}$
49. Manning's Velocity Equation: $v = (1 / n) \cdot R^{2/3} \cdot i^{1/2}$
50. Froude Number: $Fr = v / \sqrt{g \cdot D}$

PART 3: TOP 50 STANDARD VALUES & MECHANICAL CONSTANTS

1. Mass density of standard water (ρ_{water}): **1000 kg / m³**
2. Weight density of standard water (w): **9.81 kN / m³**
3. Mass density of standard mercury (ρ_{Hg}): **13600 kg / m³**
4. Specific gravity of pure mercury (S_{Hg}): **13.6**
5. Bulk Modulus of water (K_{water}): **$2.1 \cdot 10^9$ Pa**
6. Minimum piezometric tube diameter to avoid capillary error: **12 mm**
7. Capillary contact angle for pure water and clean glass (θ): **0°**
8. Capillary contact angle for mercury and clean glass (θ): **130° to 140°**
9. Standard acceleration due to gravity (g): **9.81 m / s²**
10. Standard atmospheric pressure (P_{atm}): **101.3 kPa**
11. Standard atmospheric pressure in meters of water: **10.3 m**
12. Standard atmospheric pressure in mm of mercury: **760 mm**
13. Coefficient of discharge for a standard sharp-edged orifice (C_d): **0.62**
14. Coefficient of velocity for a standard sharp-edged orifice (C_v): **0.97 to 0.98**
15. Coefficient of contraction for a standard sharp-edged orifice (C_c): **0.64**
16. Coefficient of discharge for a venturimeter (C_d): **0.95 to 0.98**
17. Coefficient of discharge for a Cipolletti weir (C_d): **0.63**
18. Side slope ratio of a Cipolletti weir: **1H : 4V**
19. Maximum theoretical efficiency for a single moving flat plate struck normally: **50%**
20. Maximum theoretical efficiency for a series of moving flat plates struck normally: **50%**
21. Maximum theoretical efficiency for a moving curved vane: **100%**
22. Maximum efficiency for power transmission through a pipe under friction: **66.7%**
23. Hydraulic mean radius (R) of an economical trapezoidal channel: **$y / 2$**
24. Slope angle of sides of a regular most economical trapezoidal channel: **60°**
25. Specific speed range for Pelton wheel with a single jet (N_s in metric): **10 to 35**
26. Specific speed range for Pelton wheel with multiple jets (N_s in metric): **35 to 60**
27. Specific speed range for Francis reaction turbine (N_s in metric): **60 to 300**
28. Specific speed range for Kaplan reaction turbine (N_s in metric): **300 to 1000**
29. Net head range for standard Pelton wheel applications (H): **> 150 m**
30. Net head range for standard Francis turbine applications (H): **30 m to 150 m**
31. Net head range for standard Kaplan turbine applications (H): **< 30 m**
32. Friction factor (f) for smooth commercial steel pipes in turbulent zones: **≈ 0.02**
33. Minor head loss coefficient for a sudden pipe contraction (K_{con}): **0.5**
34. Minor head loss coefficient for a pipe entrance (K_{ent}): **0.5**
35. Minor head loss coefficient for a pipe exit (K_{exit}): **1.0**

36. Coefficient of discharge for Borda's mouthpiece running free: **0.50**
37. Coefficient of discharge for Borda's mouthpiece running full: **0.707**
38. Coefficient of discharge for an external cylindrical mouthpiece: **0.855**
39. Angle of inclination of differential manometers to increase sensitivity: **15° to 30°**
40. Limit of y_c / d ratio for vertical thin rectangular gates: **$y_c \geq 3 \cdot d$**
41. Equivalent head conversion multiplier for mercury relative to water: **13.6**
42. Equivalent head conversion multiplier for standard oil: **0.8**
43. Standard dynamic viscosity of water at 20°C (μ): **$1.0 \cdot 10^{-3} \text{ Pa}\cdot\text{s}$**
44. Standard kinematic viscosity of water at 20°C (ν): **$1.0 \cdot 10^{-6} \text{ m}^2 / \text{s}$**
45. Standard specific speed metric base unit of power (P): **1 kW**
46. Standard specific speed metric base unit of head (H): **1 m**
47. Froude Number boundary for critical flow (Fr): **1.0**
48. Froude Number range for subcritical flow (Fr): **< 1.0**
49. Froude Number range for supercritical flow (Fr): **> 1.0**
50. Maximum efficiency of a centrifugal pump at Best Efficiency Point (BEP): **75% to 90%**

PART 4: TOP 50 SESSIONAL EXAMINER TRAPS

1. Confusing dynamic viscosity (μ) with kinematic viscosity (ν) in thermal variation trends.
2. Assuming kinematic viscosity of water is lower than mercury due to water's lower density.
3. Believing specific volume is dynamic viscosity divided by density, instead of reciprocal of density ($1 / \rho$).
4. Stating water's coefficient of compressibility is $21,000 \text{ kg / cm}^2$, which is actually Bulk Modulus (K).
5. Forgetting that compressibility is the inverse of bulk modulus ($\beta = 1 / K$).
6. Blindly adding vacuum pressure to atmospheric pressure instead of subtracting it to find absolute pressure.
7. Omitting specific gravity when converting a pressure head of non-water fluids (e.g., oil or mercury).
8. Multiplying plane submerged force by $\sin\theta$ when the surface is inclined (it remains $F = \rho \cdot g \cdot y_c \cdot A$).
9. Assuming the center of pressure on an inclined plate shifts up as the inclination angle increases.
10. Confusing the vertical component line of action with the horizontal component line of action on curved surfaces.
11. Using the actual curved gate surface area instead of the vertical projected area (A_v) to calculate F_x .
12. Swapping the center of pressure values for vertex-up vs. vertex-down vertically submerged triangular plates.
13. Using the overall vessel volume (V_{total}) instead of the submerged displaced volume (V_d) in metacentric calculations.
14. Selecting the larger longitudinal moment of inertia (I_{yy}) instead of the weaker axis (I_{xx}) in ship rolling checks.
15. Forgetting that metacentric height (GM) increases (improving stability) when BG is negative due to deep ballasting.
16. Assuming depth of flotation increases when a vessel container accelerates vertically inside an elevator.
17. Claiming the general continuity equation is based on volume conservation instead of mass conservation.
18. Applying the simplified continuity relation $A_1 \cdot v_1 = A_2 \cdot v_2$ to compressible fluids without incorporating density.
19. Confusing Euler's differential form with Bernoulli's integrated form regarding density compressibility constraints.
20. Assuming local (temporal) acceleration is convective acceleration in steady flow fields.
21. Believing HGL can never lie above a pipeline, forgetting it represents the hydraulic pressure line.
22. Assuming HGL can never rise in the direction of flow, which occurs when pipe area increases.
23. Swapping the exponents in specific speed equations: using $H^{3/4}$ for turbines or $H^{5/4}$ for pumps.
24. Calculating turbine específicos under varying head without keeping the unit quantities (N_u , Q_u , P_u) constant.
25. Forgetting that specific speed of pumps uses discharge (Q), whereas turbines use power (P).
26. Confusing energy conservation principles (Bernoulli) with momentum conservation (Euler) equations.
27. Misinterpreting velocity head as pressure head when calculating the total head along a streamline.
28. Assuming the coefficient of discharge (C_d) of Borda's mouthpiece running full is identical to running free.
29. Swapping the head exponents of notches: using $H^{3/2}$ for triangular notches or $H^{5/2}$ for rectangular ones.
30. Omitting the side slope correction factor when calculating discharge over a Cipolletti weir.
31. Using the actual top width of an economical channel instead of half the top width for sloping sides.
32. Swapping the relation between depth (y) and hydraulic radius (R) in economical rectangular sections.
33. Forgetting that specific energy decreases to a minimum at critical depth before rising again.

34. Substituting the gross head instead of net head when evaluating turbine hydraulic efficiency.
35. Confusing Pelton wheel tangential flow mechanics with Francis Mixed/Radial flow reaction pathways.
36. Assuming Kaplan turbine blades are fixed, confusing it with propeller turbines.
37. Believing a draft tube is designed to restrict tailrace flow instead of recovering velocity head.
38. Assuming cavitation occurs at high pressure zones instead of local vacuum/vapor pressure pockets.
39. Omitting pump priming procedures, assuming a centrifugal pump can self-prime from air pockets.
40. Confusing negative slip with standard positive slip in reciprocating pump diagnostics.
41. Forgetting that air vessels reduce frictional head losses rather than accelerating the stroke rate.
42. Using diameter instead of radius when calculating nozzle jet area in impact force problems.
43. Using Liters per second (L / s) directly in calculations without converting to cubic meters per second (m^3 / s).
44. Confusing the moment arm of the jet force with the gravity vector in deflected plate calculations.
45. Assuming Froude number represents the ratio of viscous forces to inertia forces, which is Reynolds number.
46. Believing a flow net can be drawn in turbulent, rotational seepage regimes.
47. Swapping the definitions of bioaccumulation (single organism) with biomagnification (trophic progression).
48. Using a factor-of-10 decimal shift on the acceleration due to gravity ($g \approx 10$ instead of 9.81).
49. Confusing the specific gravity of standard oil (0.8) with mercury (13.6) in manometer multi-fluid problems.
50. Swapping the pressure head signs in vacuum manometer equations, leading to additive atmospheric errors.

PART 5: TOP 25 "NEVER FORGET" CIVIL FLUID MECHANICS CONCEPTS

1. Mercury's kinematic viscosity (ν) is lower than water's despite its higher dynamic viscosity (μ).
2. For all standard engineering fluids, specific volume (ν) is equivalent to $1 / \rho$.
3. Capillary rise is negligible in manometers if the glass tube diameter is kept $d \geq 12$ mm.
4. Total hydrostatic force (F) on any plane submerged surface depends only on the vertical depth of its centroid.
5. Center of pressure (h^*) is always vertically below the centroid of the plane submerged surface.
6. A triangular plate submerged with its base at the surface has its center of pressure at exactly $h / 2$.
7. Horizontal hydrostatic force on a curved surface is calculated using the vertical projected area (A_v).
8. Vertical hydrostatic force on a curved surface equals the weight of the real or imaginary fluid column.
9. Ships rotate about their longitudinal axis, which represents the weakest moment of inertia.
10. Vertical acceleration inside an elevator alters effective gravity ($g_{\text{eff}} = g + a$) but leaves flotation depth unchanged.
11. Local acceleration ($\partial v / \partial t$) must equal zero under steady-state sessional flows.
12. Continuity relations for compressible flows must incorporate density ($\rho_1 \cdot A_1 \cdot v_1 = \rho_2 \cdot A_2 \cdot v_2$).
13. Bernoulli's equation is only valid for steady, incompressible, non-viscous, and irrotational flow.
14. The Total Energy Line (TEL) always slopes downwards in pipe flow unless external energy is supplied.
15. The maximum power transmission efficiency through any commercial pipeline is strictly capped at 66.7%.
16. For most economical rectangular and trapezoidal channels, the hydraulic mean radius is $R = y / 2$.
17. Specific speed of a turbine (N_s) integrates shaft power (P) and net head ($H^{5/4}$).
18. Specific speed of a pump (N_s) integrates discharge flow (Q) and net head ($H^{3/4}$).
19. Pelton wheel is an impulse turbine, whereas Francis and Kaplan are reaction turbines.
20. Draft tubes recover kinetic energy at the runner outlet by converting it to pressure head.
21. Priming is mandatory for centrifugal pumps to purge air pockets and initiate suction lift.
22. Reciprocating pumps are positive displacement machines, whereas centrifugal pumps are rotodynamic.
23. Negative slip in reciprocating pumps is caused by long suction pipes and high operating speeds.
24. Hot-wire anemometers operate strictly on thermodynamic principles to measure localized fluid velocity.
25. Flow nets remain structurally unchanged unless the physical boundary conditions of the seepage field are modified.